HYDRA Documentation

Installation Guide SQL Server 2019

for HYDRA MW4.0pe

Version 1.0.23049

Last changed on: 02.09.2020

Contents

1 Installation SQL Server 2019 SE

Attention:

To perform the following procedures advanced knowledge of your server operating system, the intended database software and common IT systems is required.
These procedures are performed on your own responsibility.
MPDV Mikrolab GmbH is not liable for any loss of or destruction of data.
In case of doubt place an order with MPDV Mikrolab GmbH to perform these tasks for you.

This manual explains how to install Microsoft SQL Server 2019 Standard Edition for use with MPDVs Manufacturing Integration Platform (**MIP**) based products like MIP 1.1, HYDRA MW4.0pe or FEDRA 1.1.

Other than MIP the applications HYDRA and FEDRA allow for a multi system installation (multiple HYDRA or FEDRA systems on the same (application) server).
For multi system installations every HYDRA or FEDRA system needs its own database.
Name your databases **mip1**, **mip2**, **mip3**, etc.
With multi system installations it is strongly recommended that every database has its own database instance available, e.g.: **MIPMS1**, **MIPMS2**, **MIPMS3**, etc.

Attention: Do not run these installation procedures on a computer where MIP based products like MIP 1.1, HYDRA MW4.0pe or FEDRA 1.1 are already running.
By doing so, you might destroy your existing MIP based system.
Please be extremely cautious if you decide to do it anyway.

1.1. Installation Requirements

- Server hardware and software according to the recommendations of MPDV Mikrolab GmbH for the respective MIP based product.
  See manuals: **HW_SW_GUIDE.pdf**

- English or German Windows operating system according to the recommendations of MPDV Mikrolab GmbH.

- Server configuration according to MPDVs server configuration manuals for the respective MIP based product, e.g.:
  "**Preparation Guide for Windows Server 2016**" (e.g.: PreInstGuide_MW40_WIN2016.pdf)
  "**Preparation Guide for Windows Server 2019**" (e.g.: PreInstGuide_MW40_WIN2019.pdf)

- Installation DVD or installation files for the required edition of "**SQL Server 2019**", e.g. for Standard Edition, e.g.:
  English: **en_sql_server_2019_standard_x64_dvd_c7d70add.iso**
  German: **de_sql_server_2019_standard_x64_dvd_2cd4ee21.iso**

- Installation files for the Microsoft **SQL Server Management Studio (SSMS)** matching the language of your operating system, e.g.:
  English: **SSMS-Setup-ENU.exe**
  German: **SSMS-Setup-DEU.exe**
  (SQL Server Management Studio is no longer included in the installation DVD of the SQL Server!)

- The language versions of the operating system and the database must match!

- Service Pack (**SP**) and Cumulative Update Package (**CP**) installation files for SQL Server 2019 mandatory for the respective MIP based product:
  Service Pack (**SP**): currently there is no SP installation mandatory for MIP, HYDRA or FEDRA (only if the required SP is not already included in the original installation file or DVD)
  Cumulative Update Package (**CP**): currently there is no CP installation mandatory for MIP, HYDRA or FEDRA

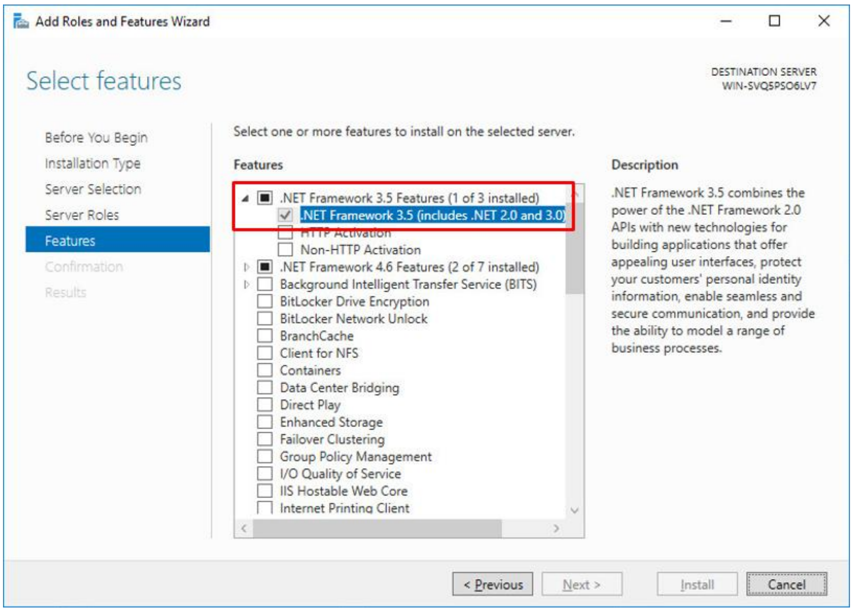
- Minimum hard disk configuration:
  Partition **C:** with a minimum of **100 GB** disk space available and file system **NTFS**.
  Partition **D:** with a minimum of **600 GB** disk space available and file system **NTFS**.
  For more details please see the server configuration manual.

- For better performance it is recommended to have at least two hard disks available:
  Disk 1 with partition **C:** (≥ 100 GB) and **D:** (≥ 200 GB) (for Windows and applications)
  Disk 2 with partition **E:** (≥ 400 GB) (for database files)
  For more details please see the server configuration manual.

- The Windows desktop resolution should be set to **1920 x 1080** at least to **1280 x 1024** (see server configuration manual).

- Local Windows user **mipadm** (see server configuration manual).
  The user **mipadm** must be member of the local group **Administrators**.

- Activated Feature: **.NET Framework 3.5**
  -Button → Server Manager → Manage → Add Roles and Features

  To add this feature the directory x:\sources\sxs from the Windows installation DVD is needed! (see server configuration manual)

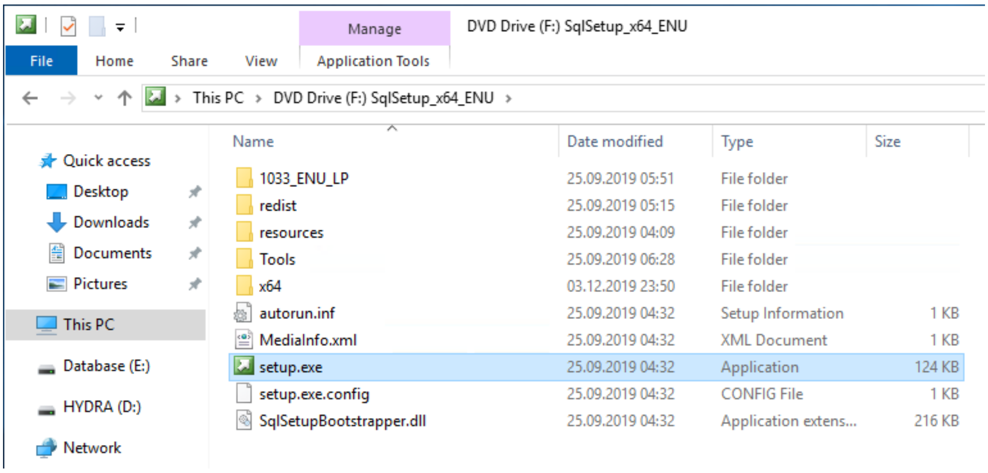
- With **separate MIP based application and database servers** it is mandatory to install additional software on the **application server**.
  Please see chapter "[1.7. Separate Database and Application Server]".

1.2. Installing SQL Server 2019

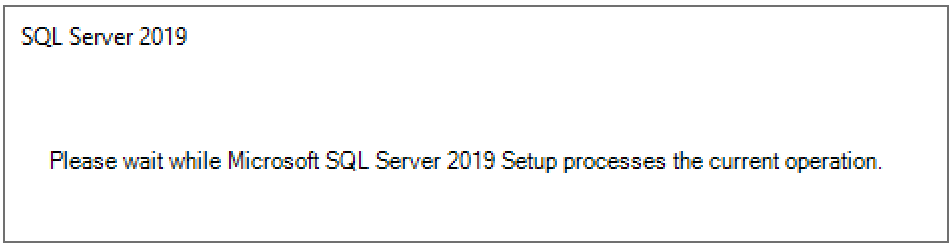
Login as local user **mipadm**.

Insert the SQL Server installation disc or mount the installation file, e.g.:
en_sql_server_2019_standard_x64_dvd_c7d70add.iso

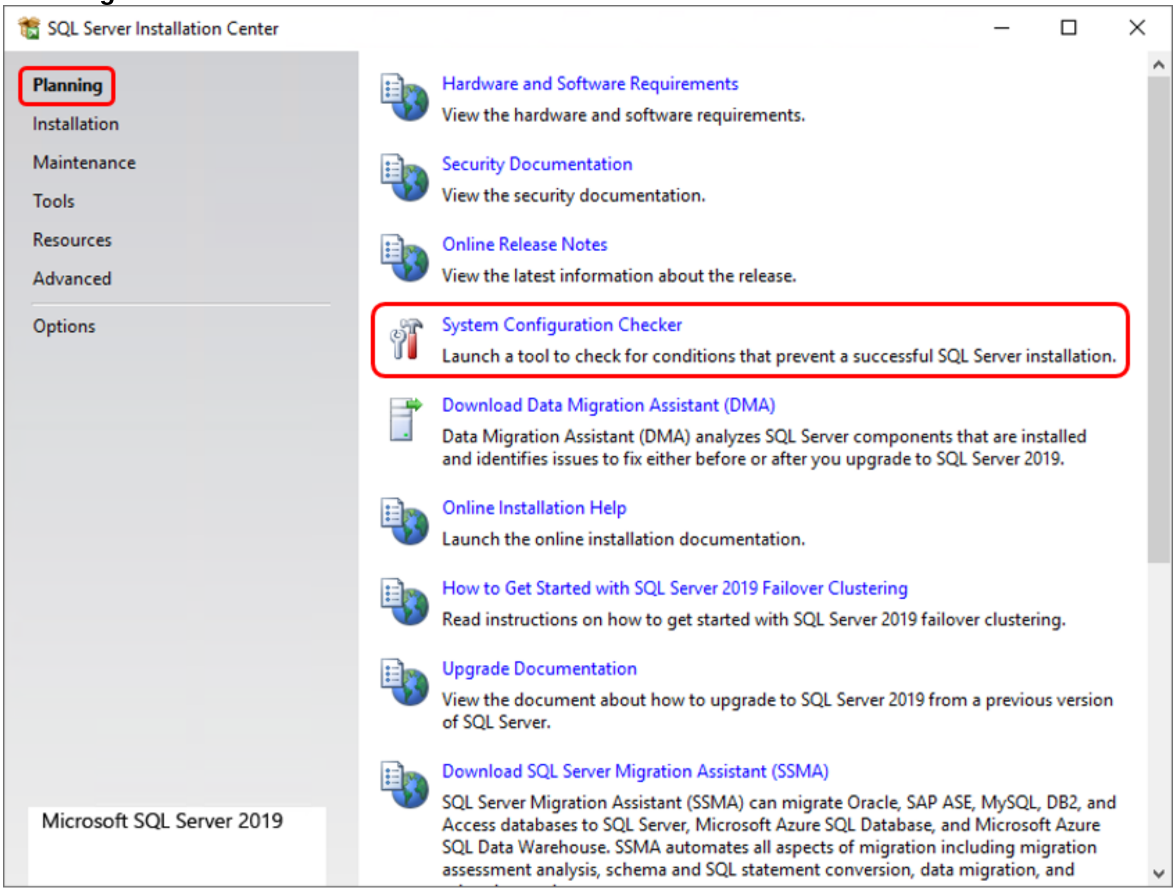
If the installation is not starting automatically start it manually by double clicking **setup.exe**.



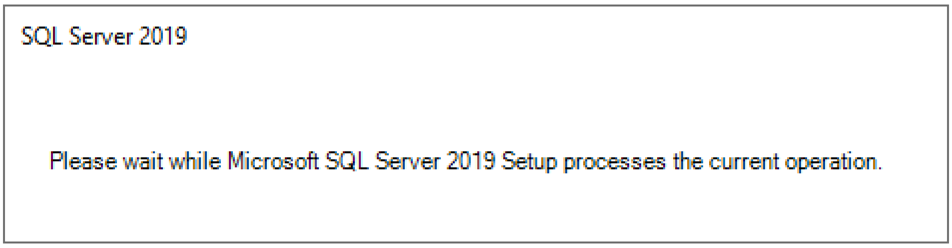
Please proceed with the installation as described below:

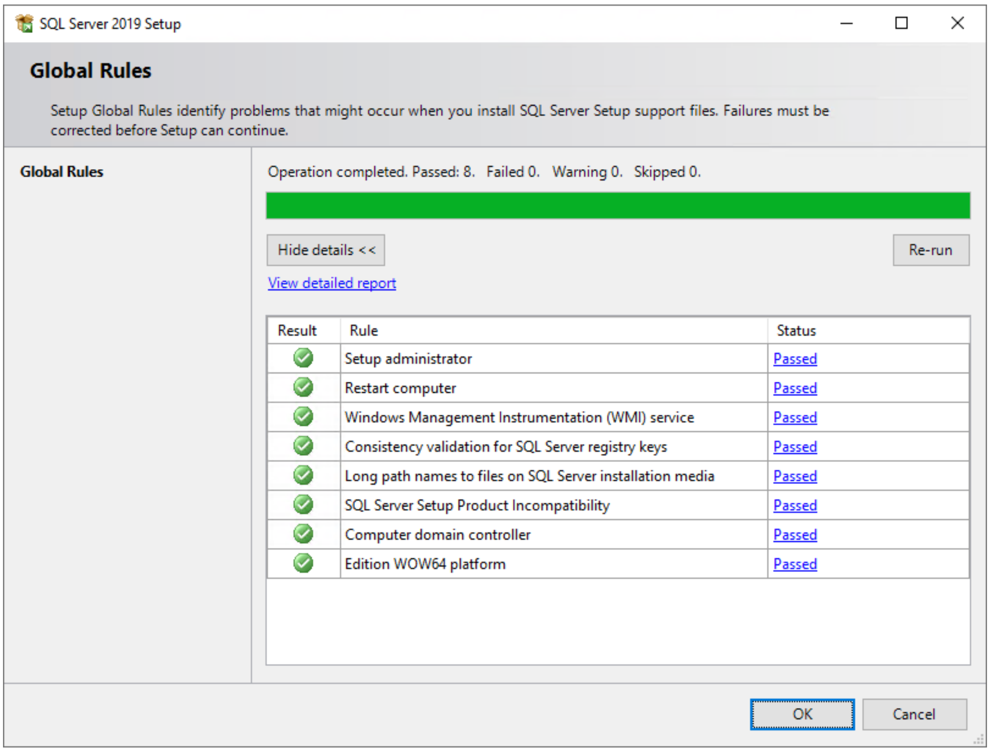


Planning



Select: System Configuration Checker

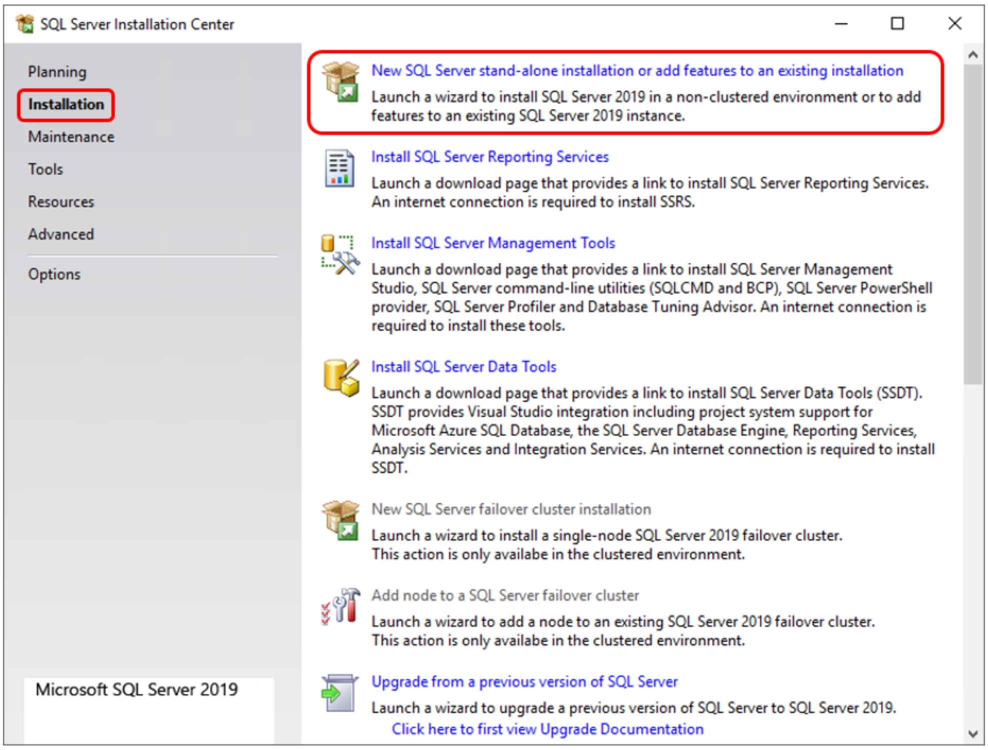




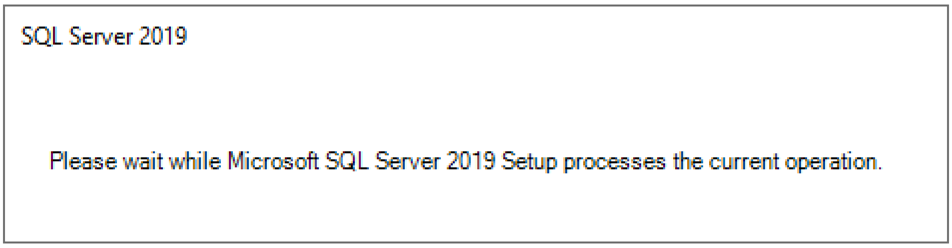
There should be no warnings or errors

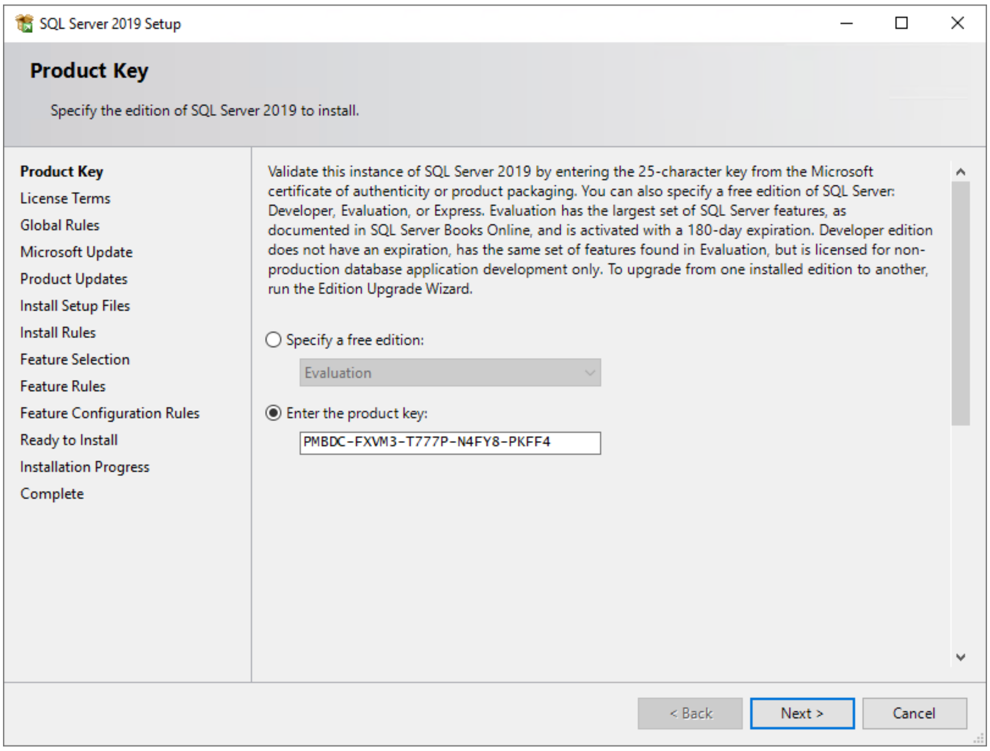
OK

Installation



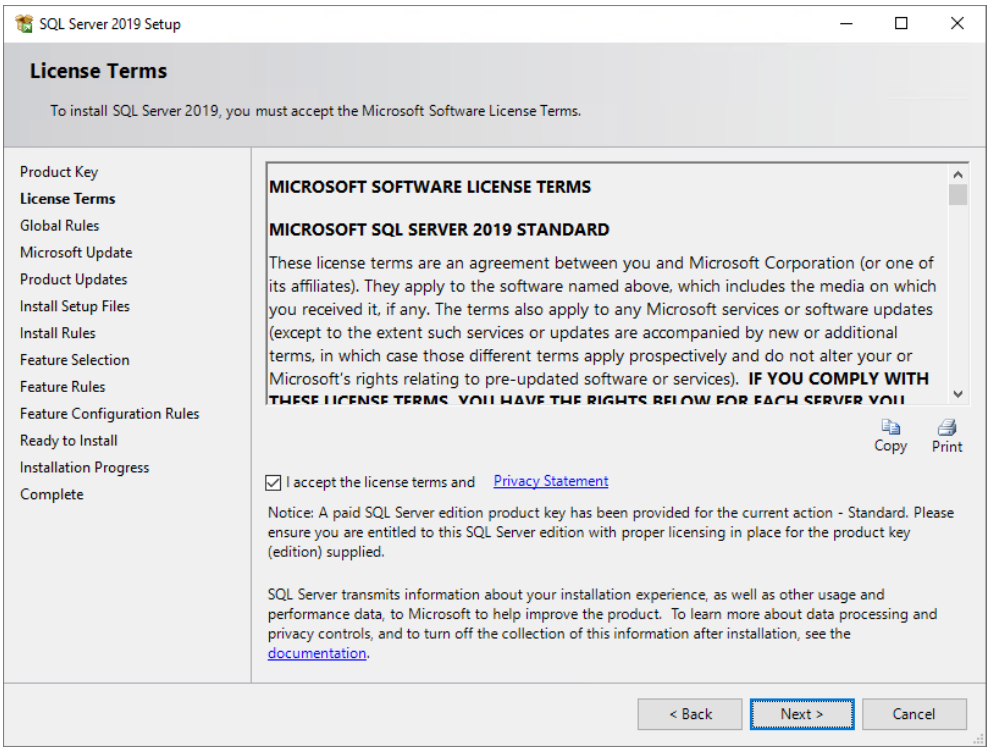
Select: New SQL Server stand-alone installation or add features to an existing installation





Enter your product key (if necessary)

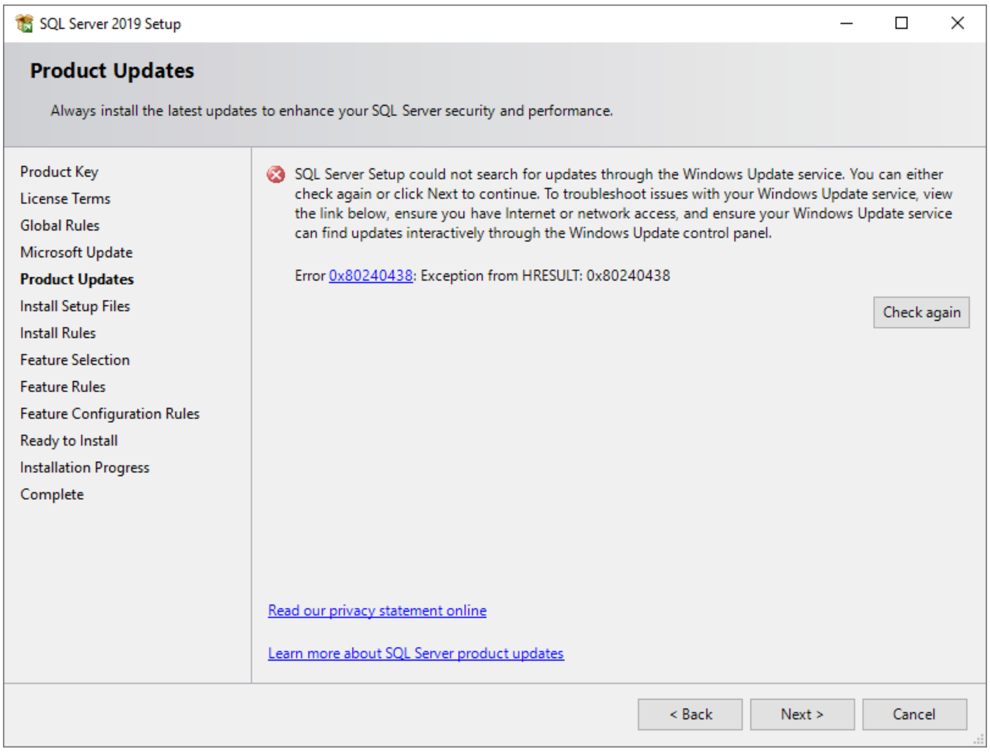
Next



Activate: **I accept the license terms**

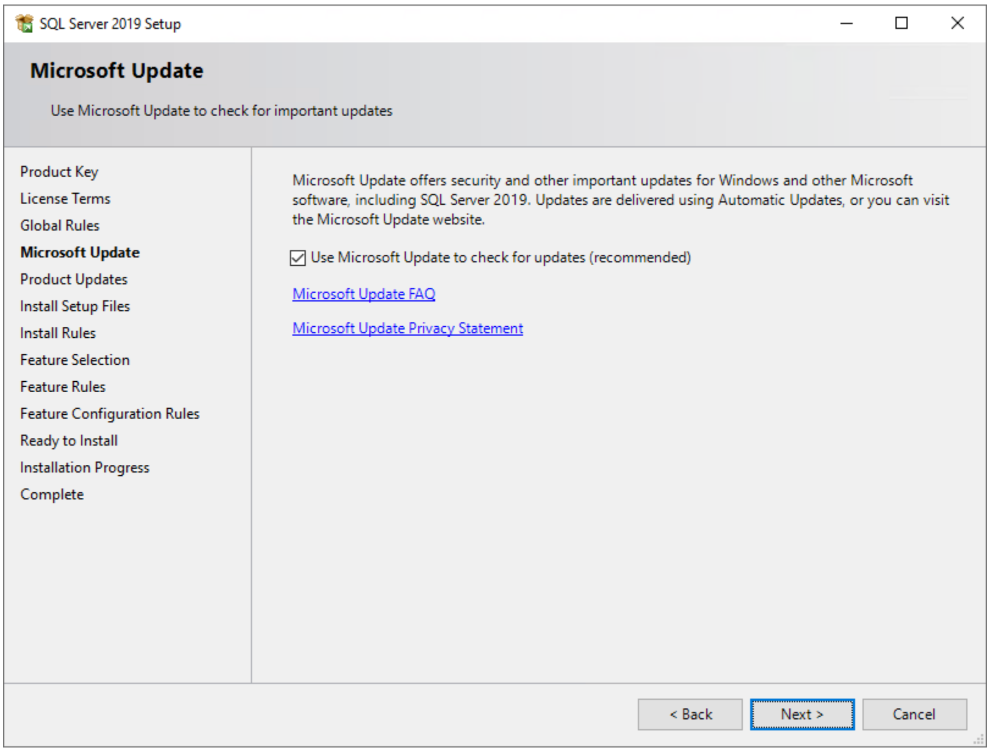
Next

If your server has no access to the internet you will see a message like this:



Next

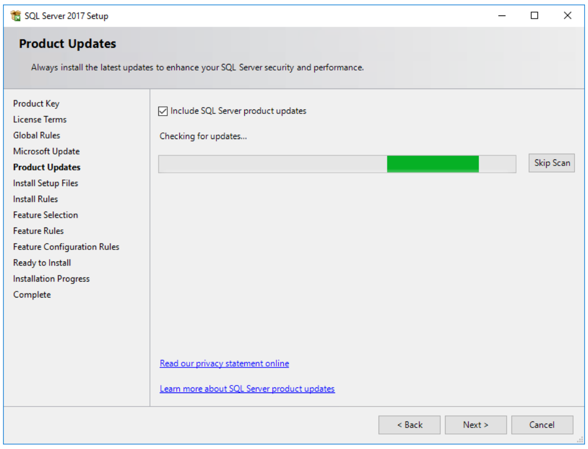
If your server has access to the internet you will see the following message:

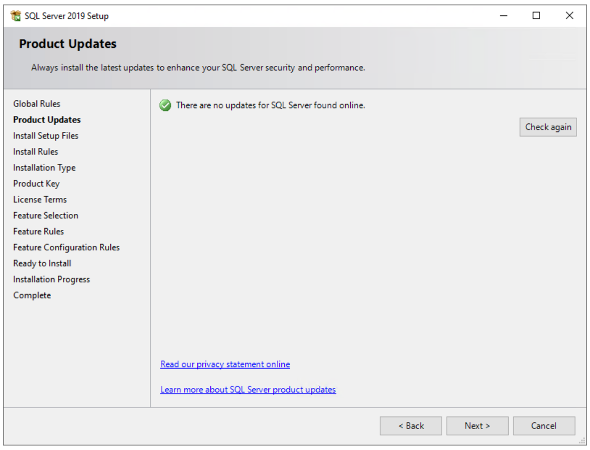


Activate: Use Microsoft Update to check for updates (recommended)

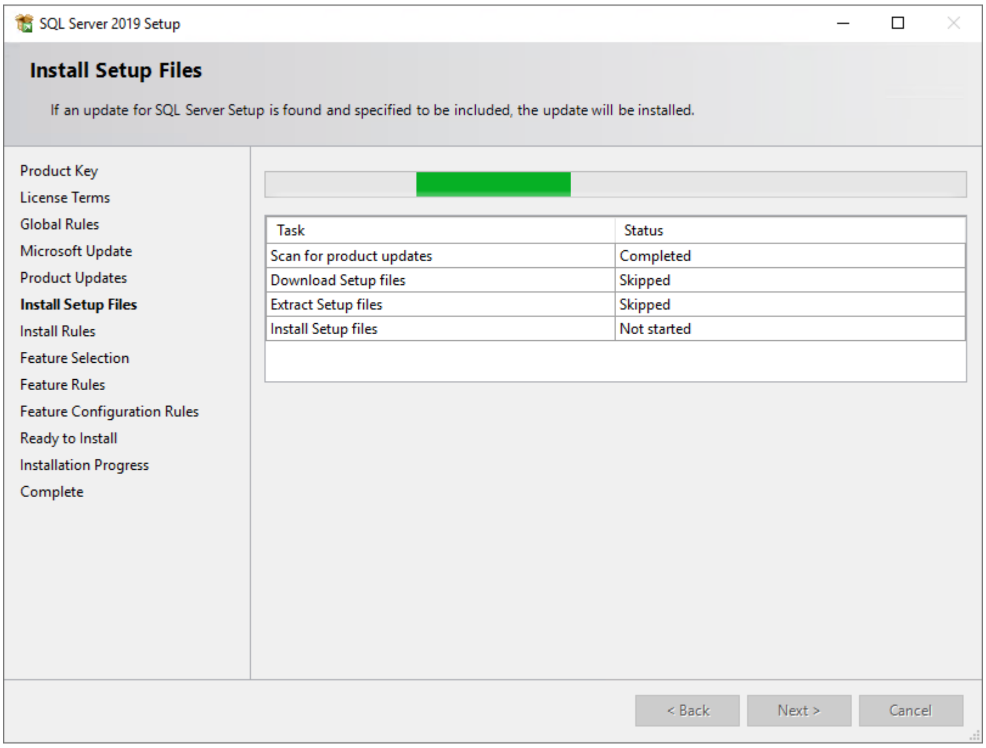
Next

If there are SQL Server product updates available you might want to include them in your installation:

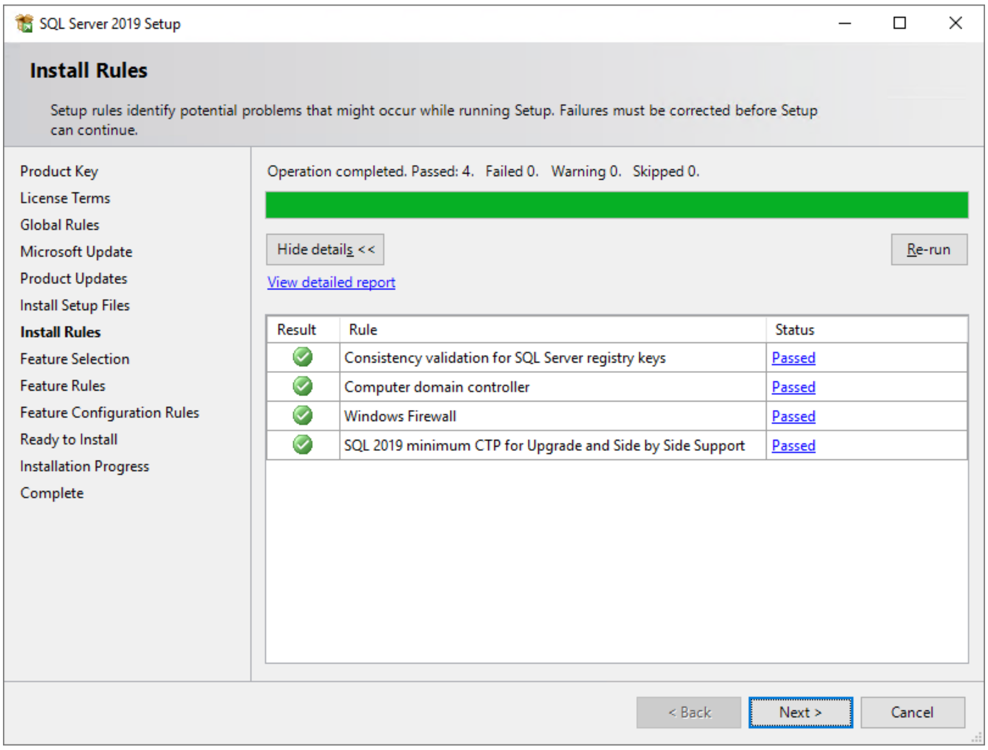




Next

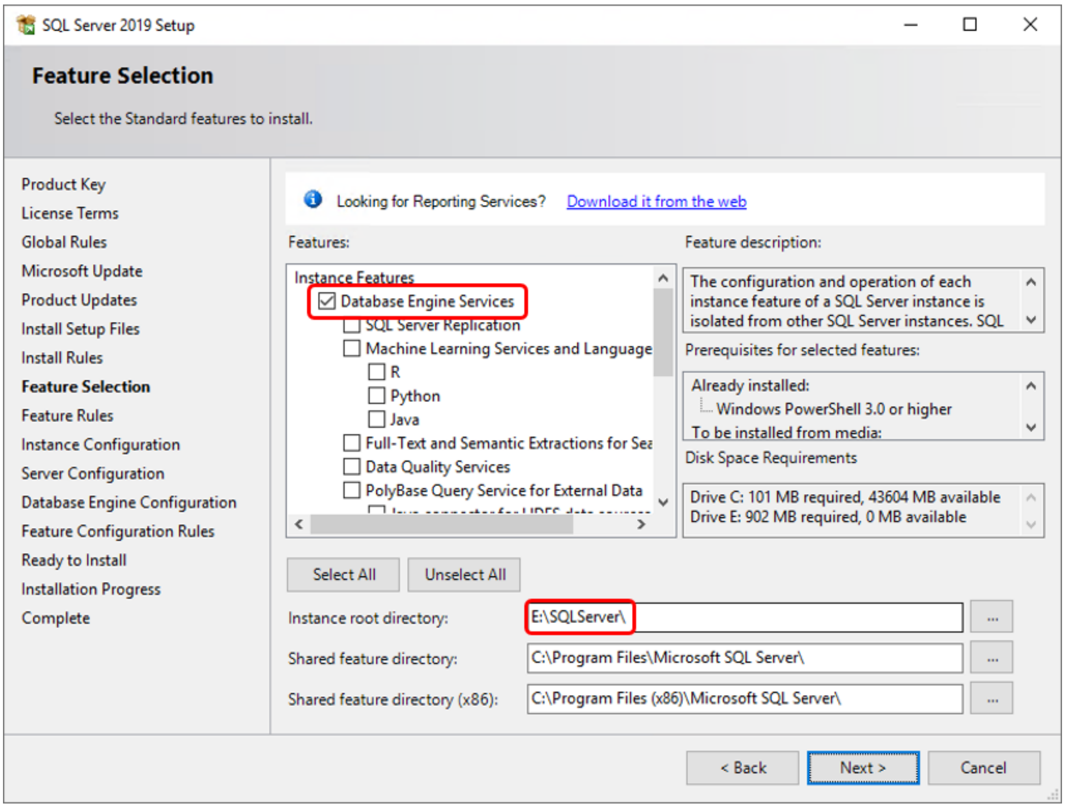


There should be no errors or warnings:



Possible warnings for "Windows Firewall" which is supposedly active but actually isn't, can be ignored.

Next

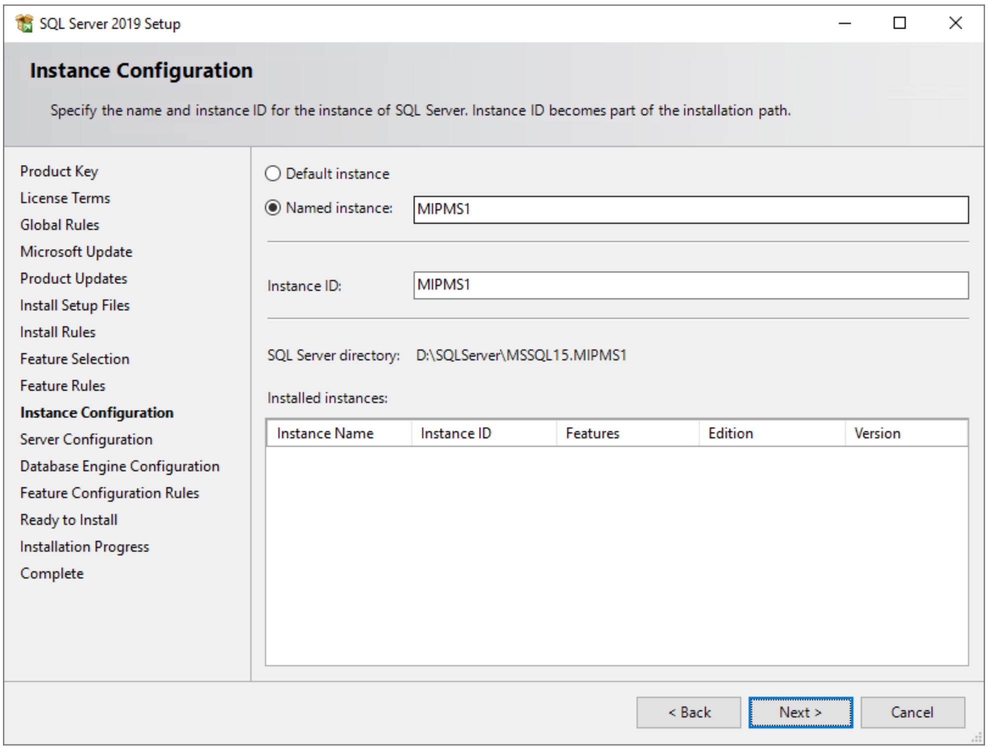


Select the features to be installed: **Database Engine Services**

Select Instance root directory, e.g.: **D:\SQLServer** (when using 1 hard disk)

E:\SQLServer (when using 2 or more hard disks)

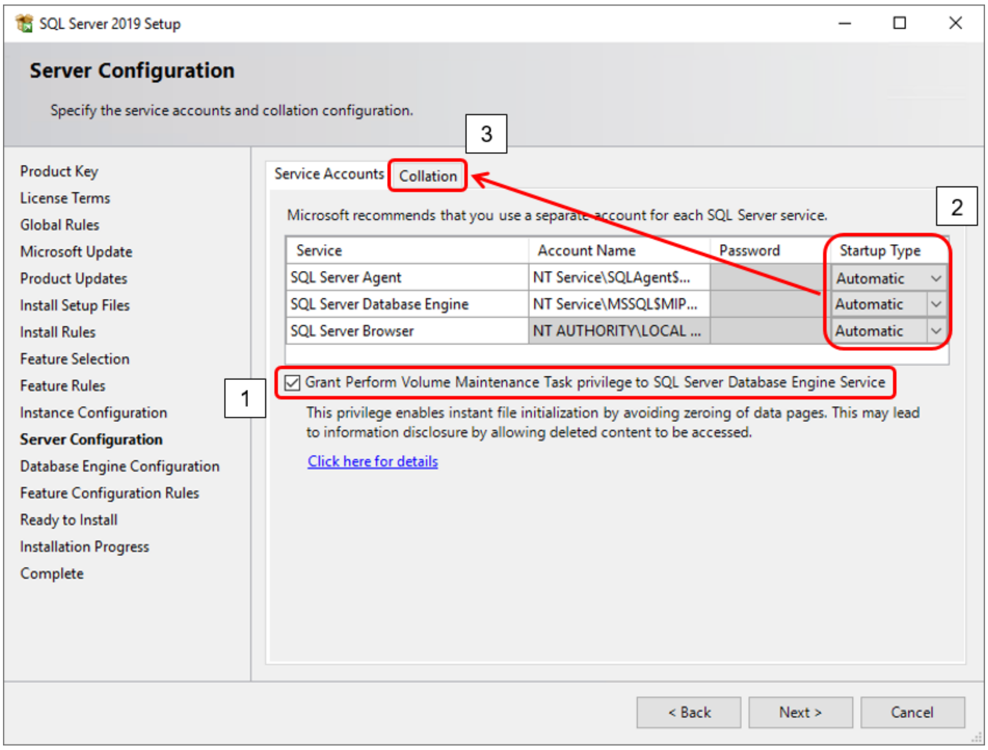
Next



Named instance: **MIPMS1**

Instance ID: **MIPMS1**

Next

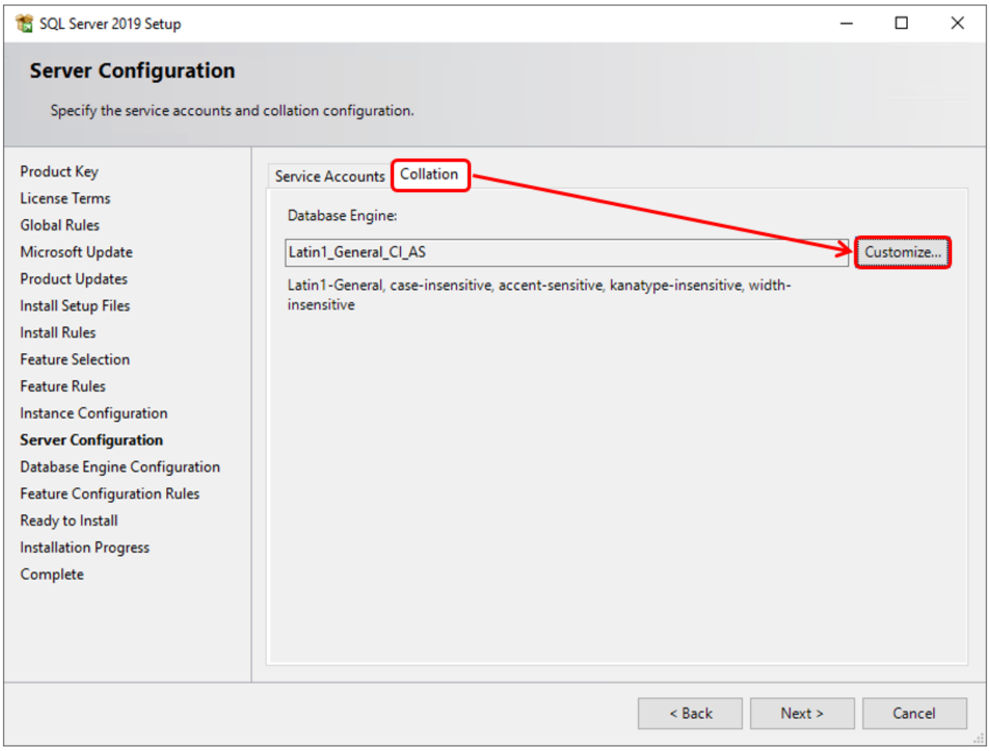


Activate: **Grant Perform Volume Maintenance Task privilege to SQL Server …**

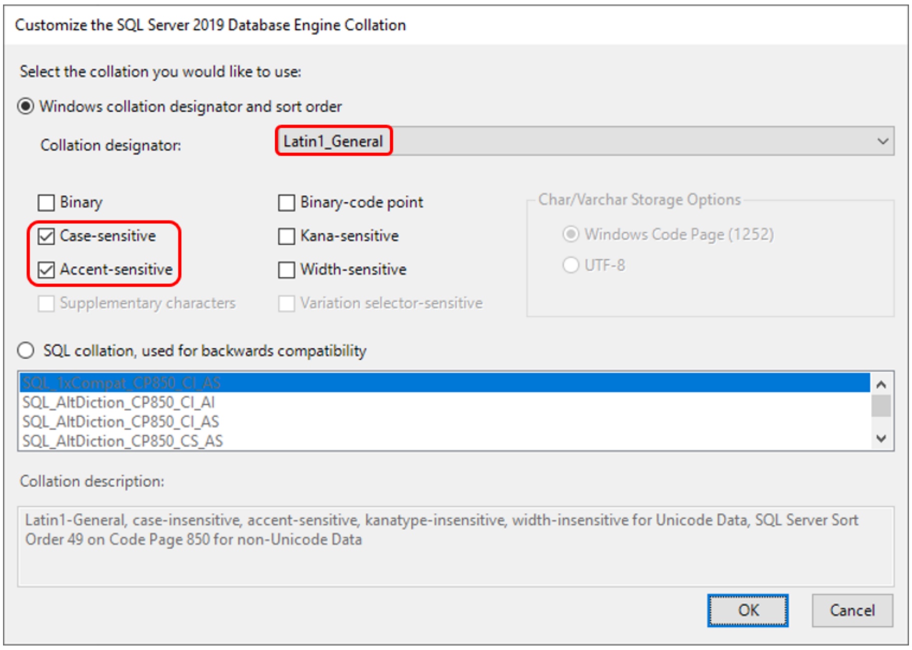
Startup Type: **Automatic** **(All services must start automatically!)**

Select Tab: **Collation**

Collation



Customize

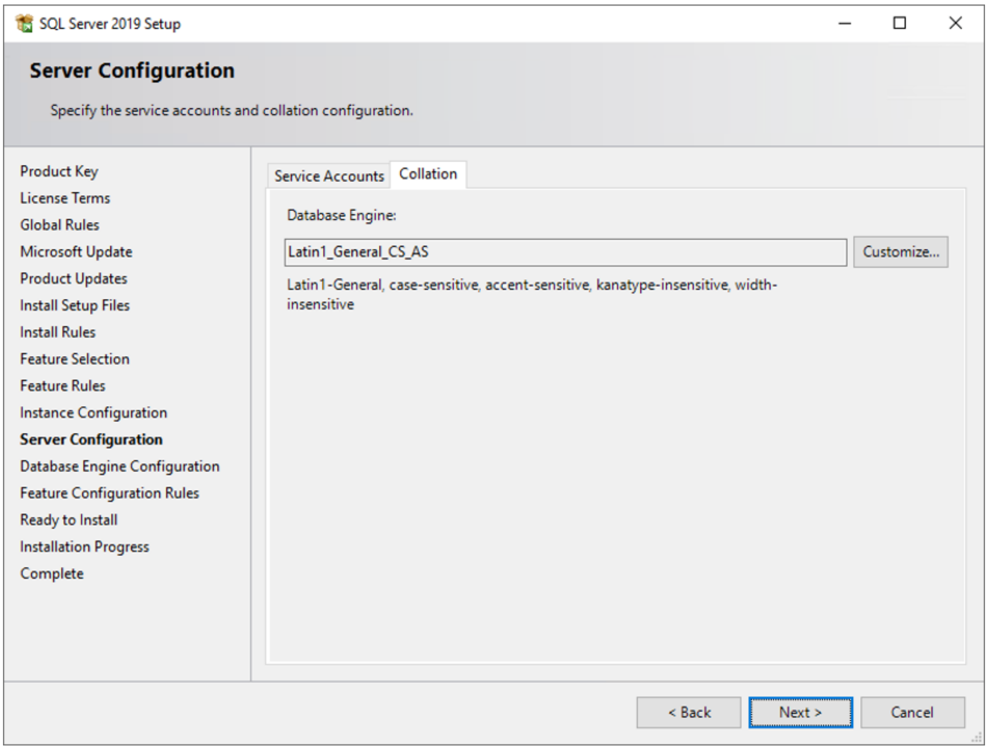


Collation designator: Latin1_General

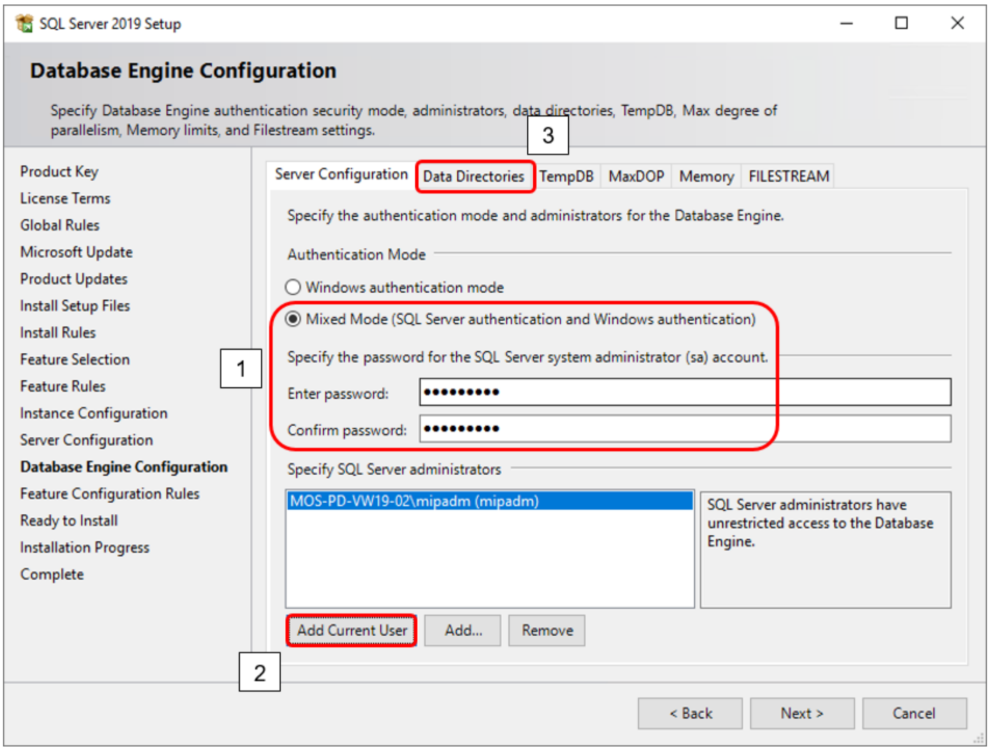
Activate: Case-sensitive

Activate: Accent-sensitive

OK



Next



Activate: **Mixed Mode** **(mandatory for MIP based products!)**

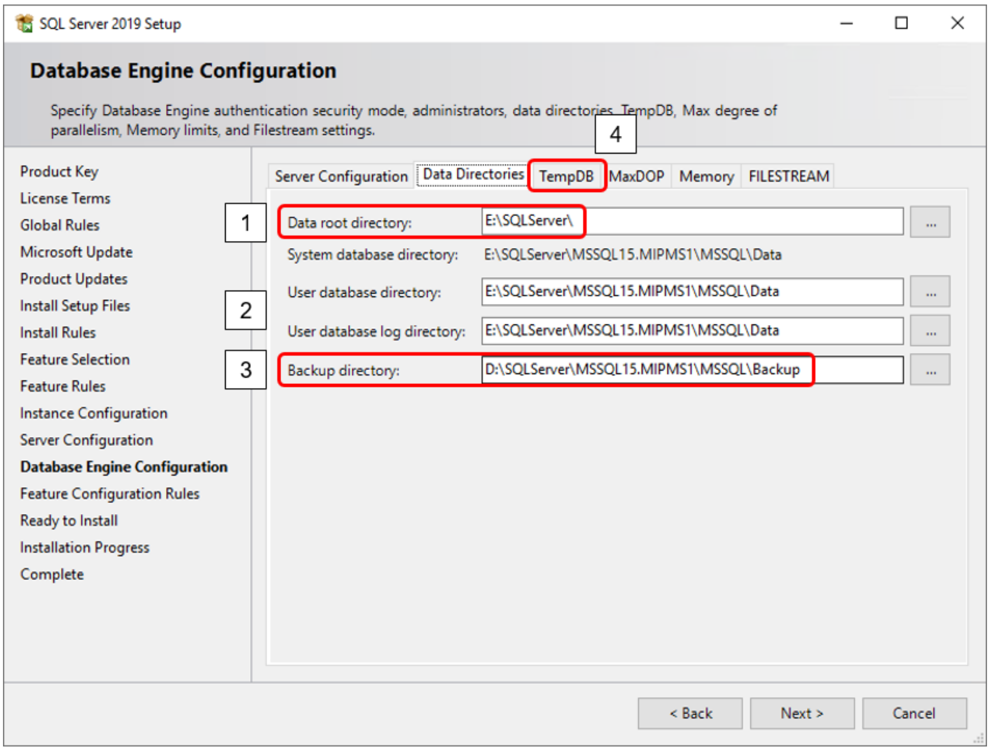
Enter password: **<SECRET sa PASSWORD>**

Confirm password: **<SECRET sa PASSWORD>** (password should be disclosed to MPDV)

Click on Button: **Add Current User**

Select Tab: **Data Directories**

Data Directories



Data root directory:	**D:\SQLServer**	(when using 1 hard disk)
	E:\SQLServer	(when using 2 or more hard disks)

User database directory: **change only if necessary**

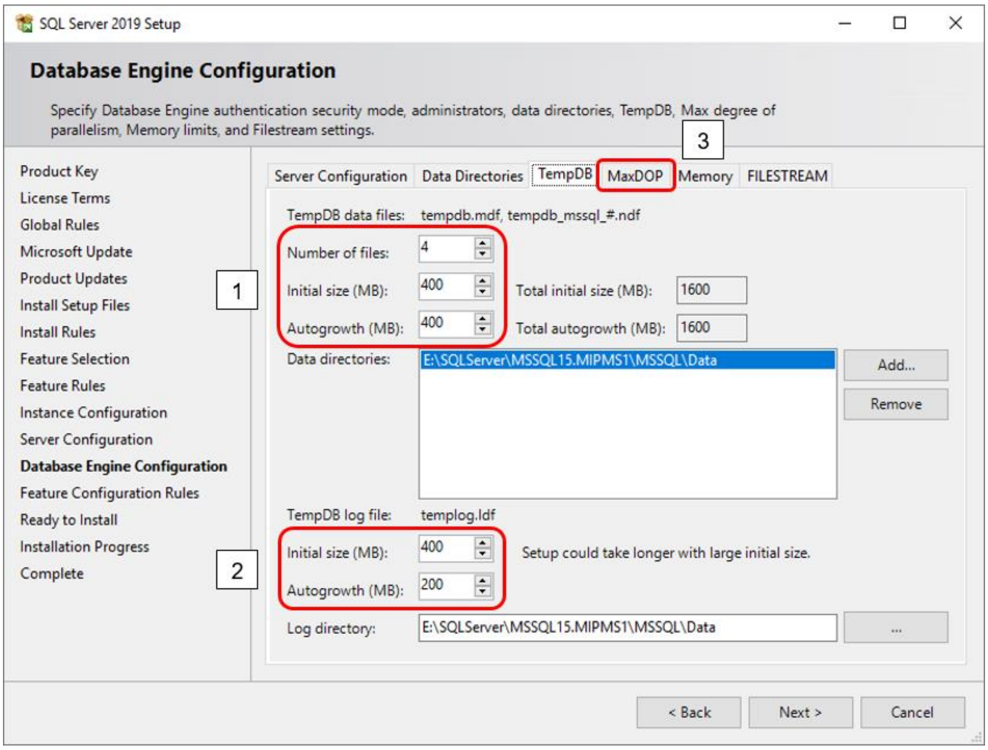
User database log dir.: **change only if necessary**

Backup directory: **D:\SQLServer\MSSQL15.MIPMS1\MSSQL\Backup**

(The backup directory should not be on the same disk as database data and log files!)

Select Tab: **TempDB**

TempDB



TempDB data files:

Number of files: 4 (Default, do not use less than 4 files!)

Initial size (MB): **400**

Autogrowth (MB): **400**

Data directories: **change only if necessary,**
e.g. to place TempDB data files on separate disks

TempDB log file:

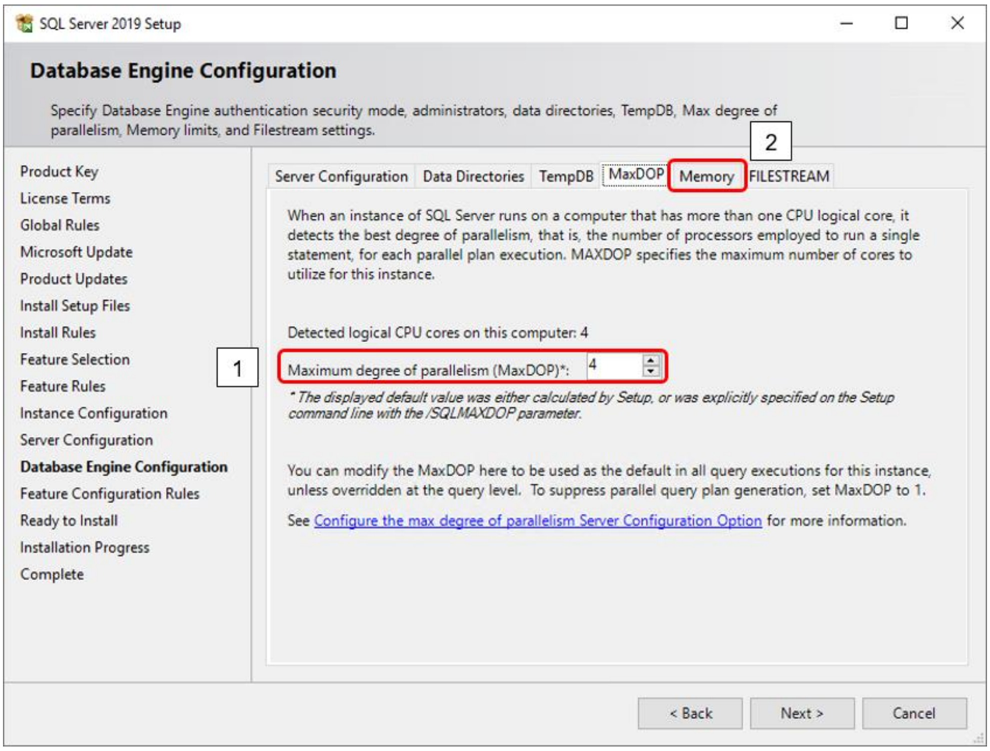
Initial size (MB): **400**

Autogrowth (MB): **200**

Log directory: **change only if necessary,**
e.g. to place TempDB log file on a separate disk

Select Tab": **MaxDOP**

MaxDOP

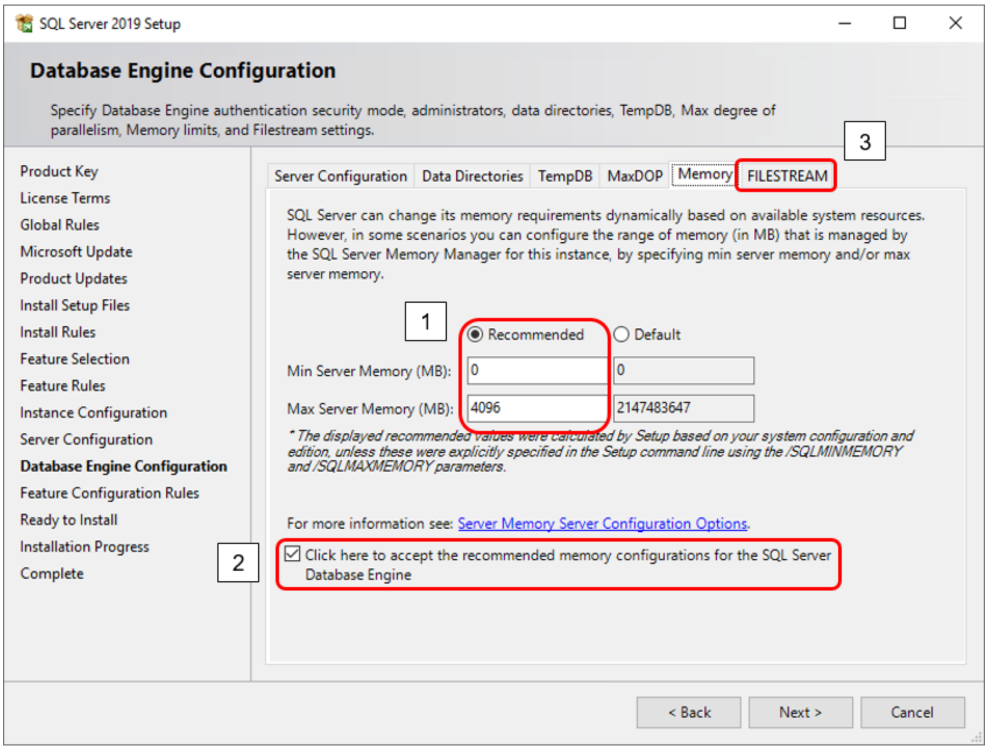


Check the setting for "**Maximum degree of parallelism (MaxDOP)**" which was calculated by Setup.

Make sure it is set to a value of at least 4

If the calculated value is higher than 4 you may leave it that way.

Select Tab: **Memory**



Change radio button from Default to Recommended:

Min Server Memory (MB): **0**

Max Server Memory (MB): **adjust according to the statement below**

(Note that the displayed recommended value does not consider the following circumstances and might need to be adjusted accordingly)

Depending on the total amount of RAM memory available on the server and the system requirements by the MIP based product please set the size for “Max Server Memory (MB)” so that as much memory as possible will be used by the database instance.

The size of “Max Server Memory (MB)” will have direct influence on the performance of the MIP based application because with bigger sizes more data can be cached by the database.

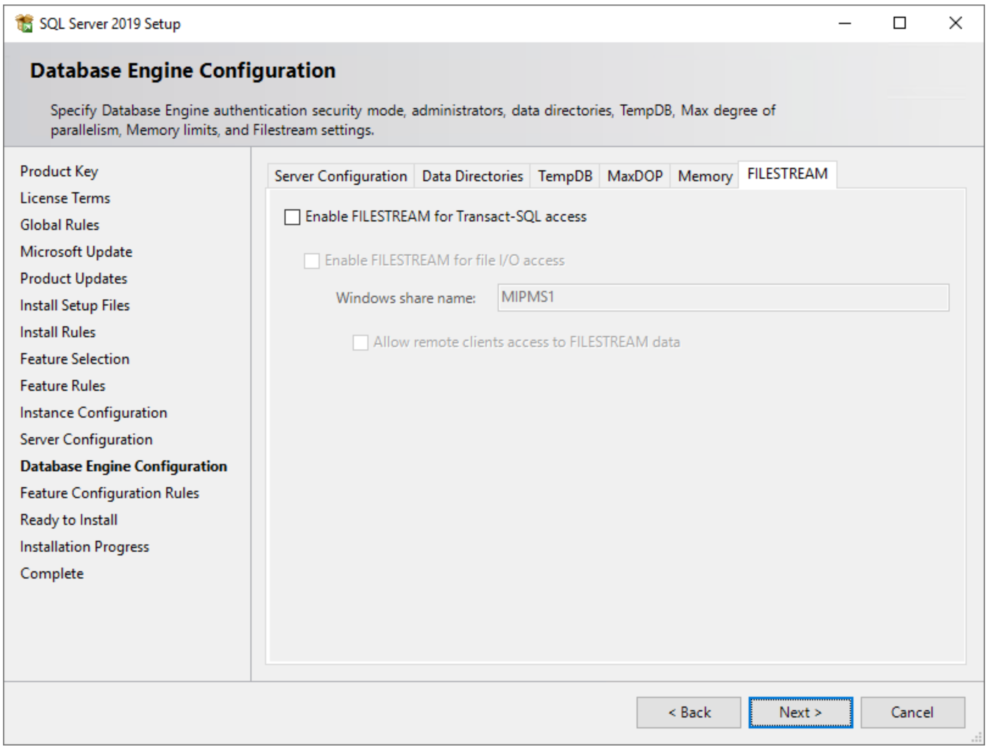
As long as it is ensured that there will be enough RAM left for the requirements of the server’s operating system (~2GB), the MIP based product services (depending on the system size ~1,5-8GB per system), the MIP WSP service (depending on the system size ~2-12GB per system) and other database instances installed on the same server you should use as much memory for “Maximum Server Memory” as possible.

Make sure that the available server memory (RAM) is used to full capacity without forcing your server into swapping.

Activate the checkbox to accept the recommended memory configurations for the SQL Server Database Engine

Select Tab: **FILESTREAM**

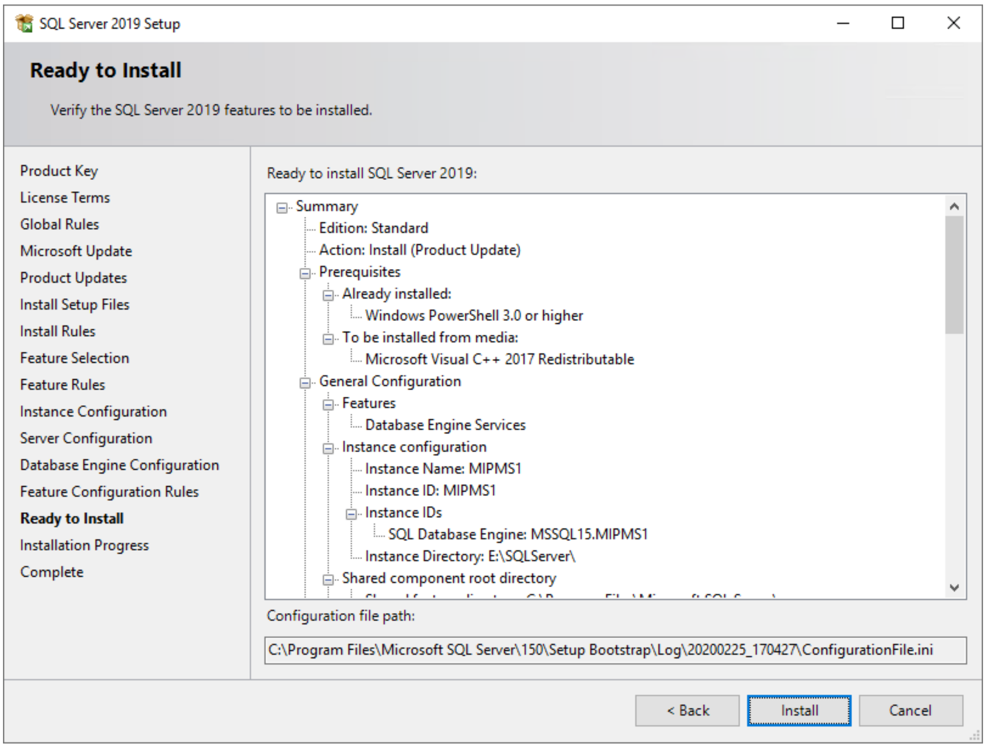
FILESTREAM



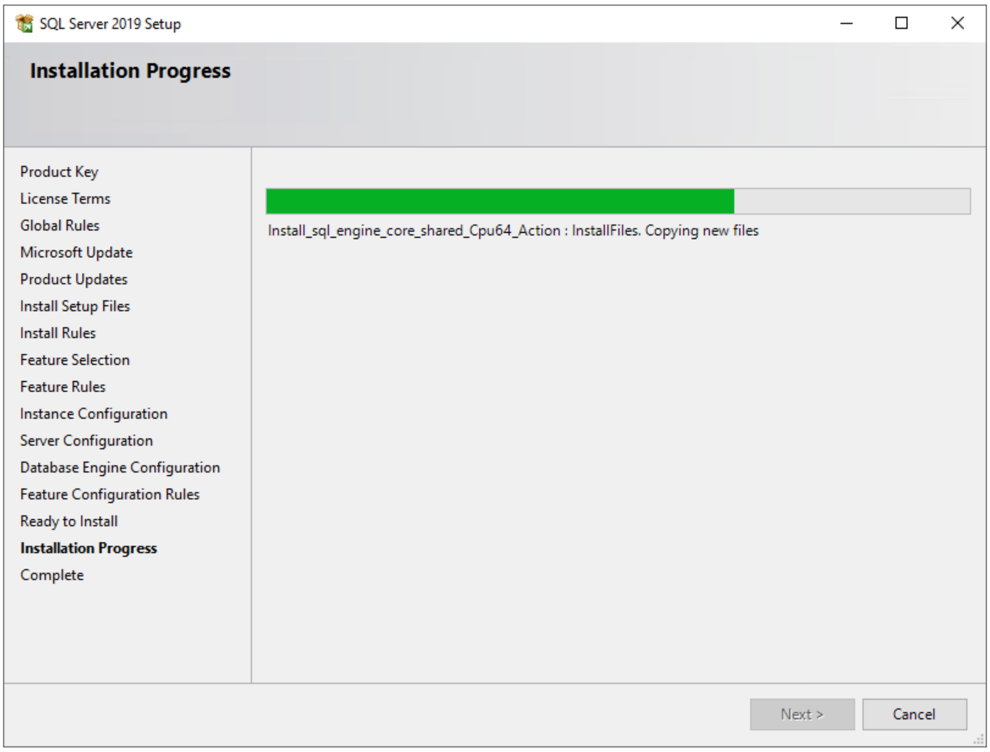
no changes necessary

Next

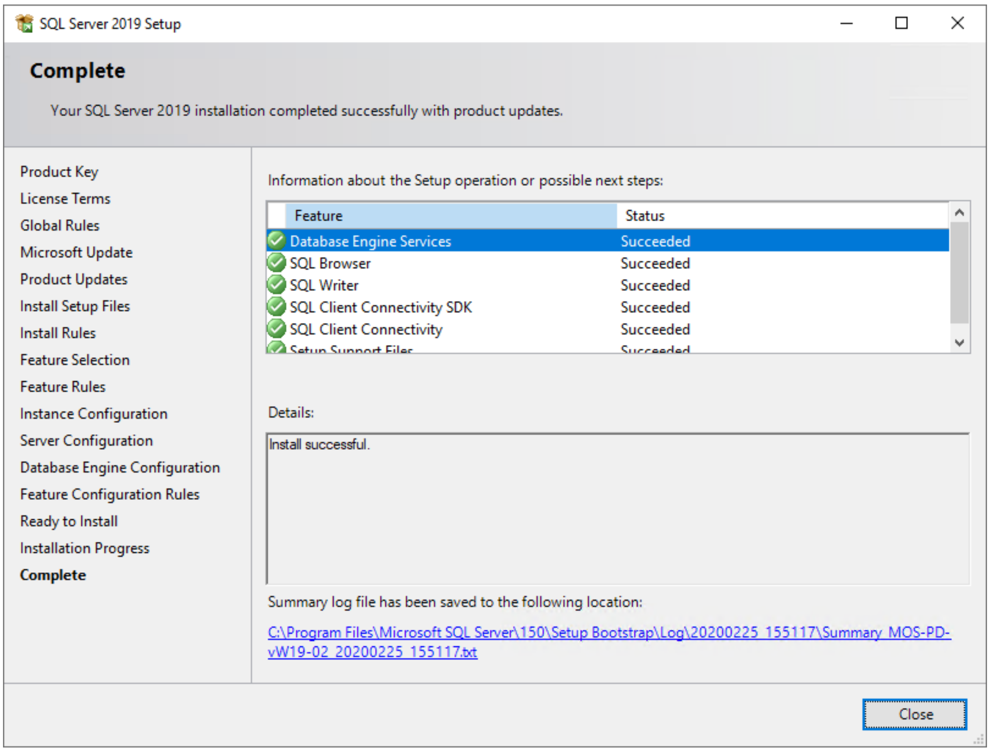
Verify the SQL Server features to be installed



Install

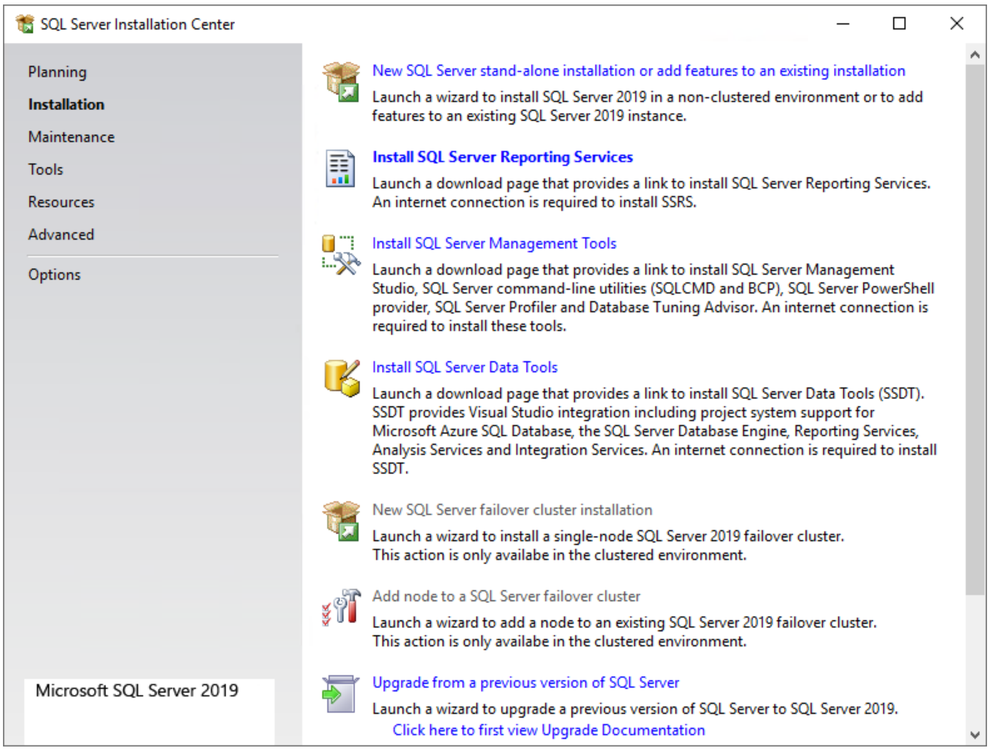


Make sure the installation succeeded successfully



Close

Close this Window



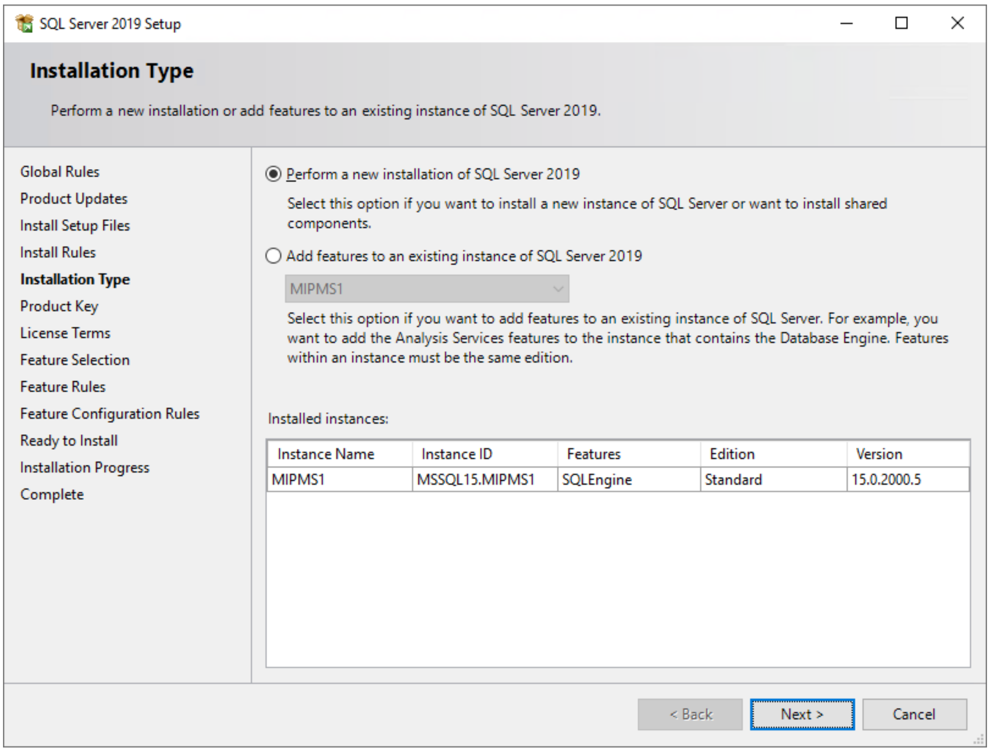
For HYDRA or FEDRA multi system installations repeat the installation procedures to add additional database instances (see also next chapter).

1.2.1. Installing additional Database Instance

Other than MIP the applications HYDRA and FEDRA allow for a multi system installation (multiple HYDRA or FEDRA systems on the same (application) server).
For multi system installations every HYDRA or FEDRA system needs its own database.
Name your databases **mip1**, **mip2**, **mip3**, etc.
With multi system installations it is recommended that every database has its own database instance available, e.g.: MIPMS1, MIPMS2, MIPMS3, etc.

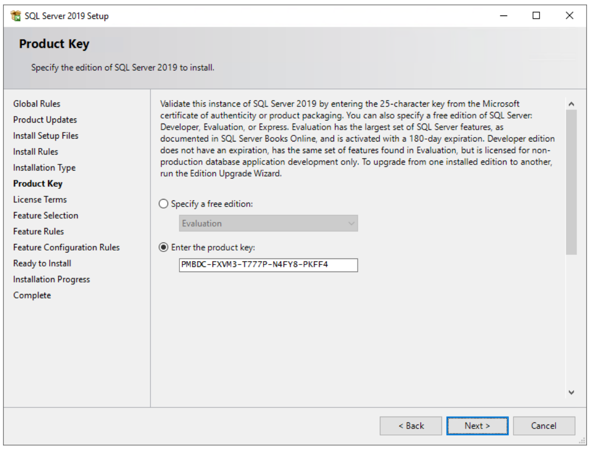
Start an installation as described in chapter "1.2. Installing SQL Server 2019"

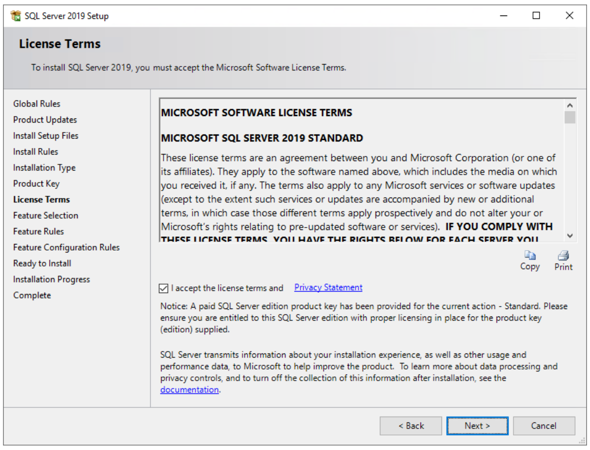
To install a new instance of SQL Server please proceed as follows:



Perform a new installation of SQL Server 2019

Next



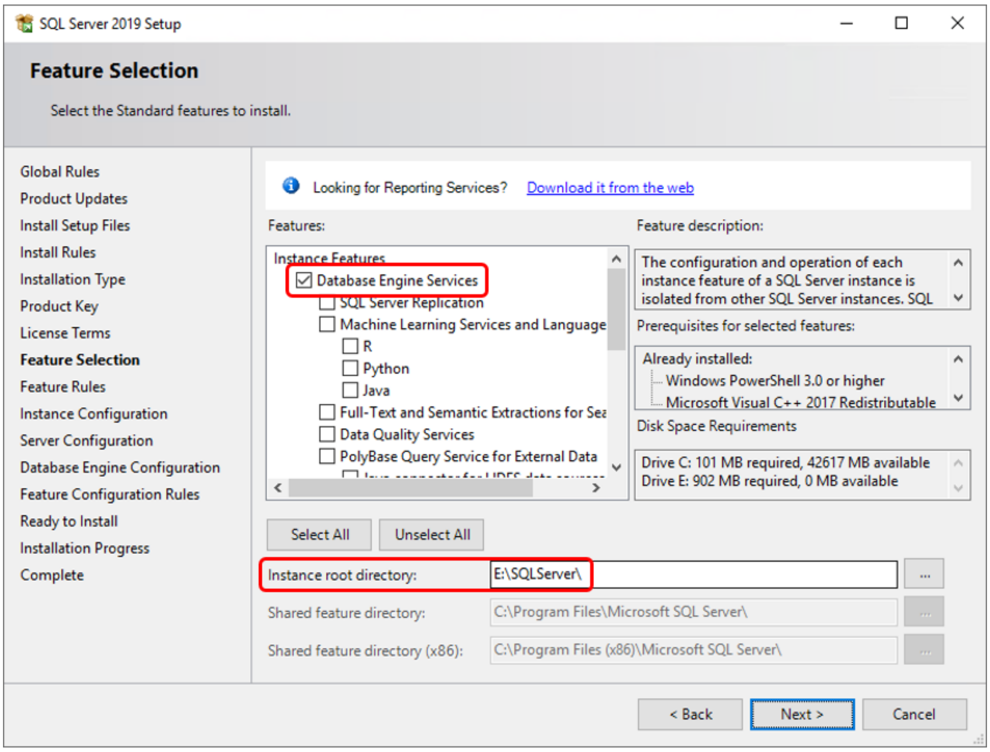


Enter product key (if necessary)

Next

Activate: I accept the license terms

Next

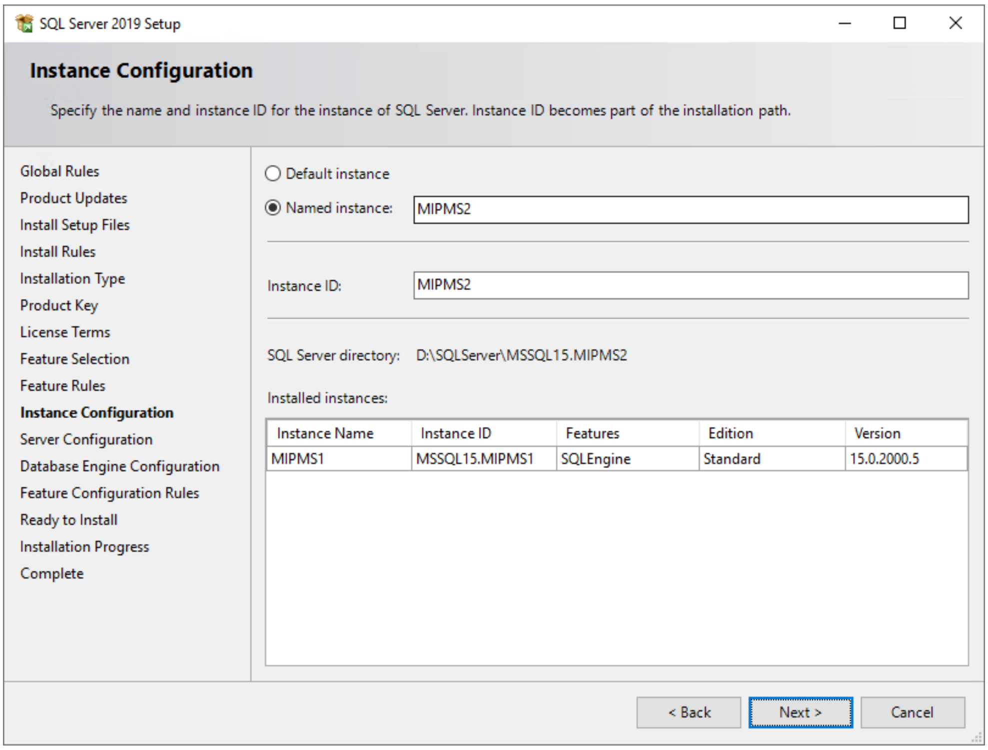


Select the features to be installed: **Database Engine Services**

Select Instance root directory, e.g.: **D:\SQLServer** (when using 1 hard disk)

E:\SQLServer (when using 2 or more hard disks)

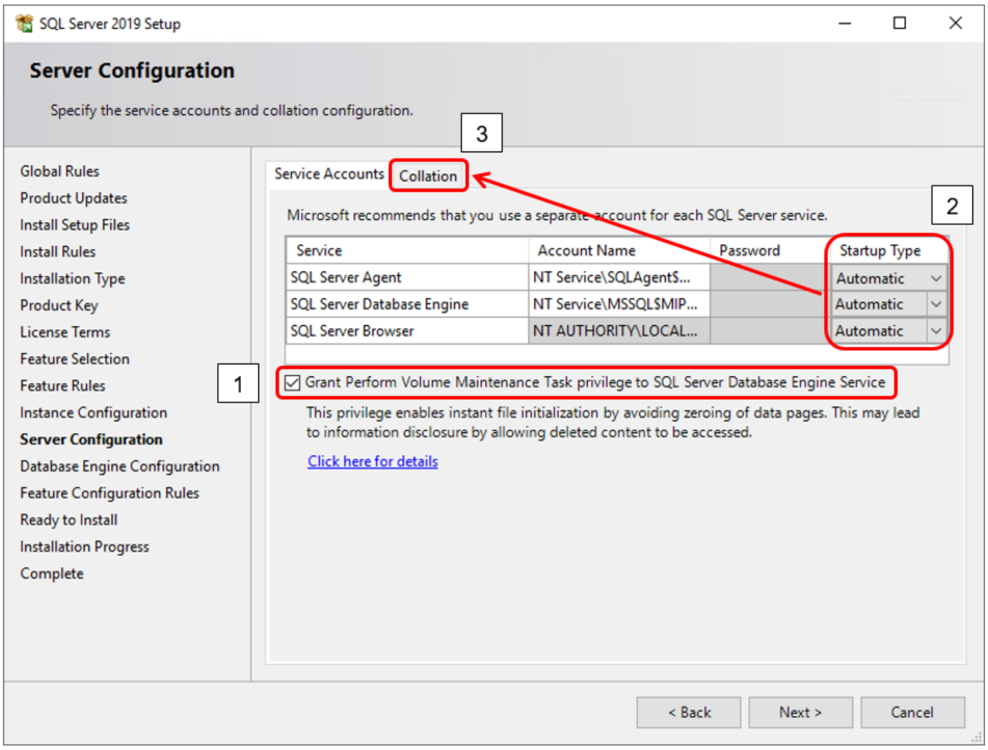
Next



Named instance: **MIPMS2, MIPMS3, MIPMS4, etc.**

Instance ID: **MIPMS2, MIPMS3, MIPMS4, etc.**

Next

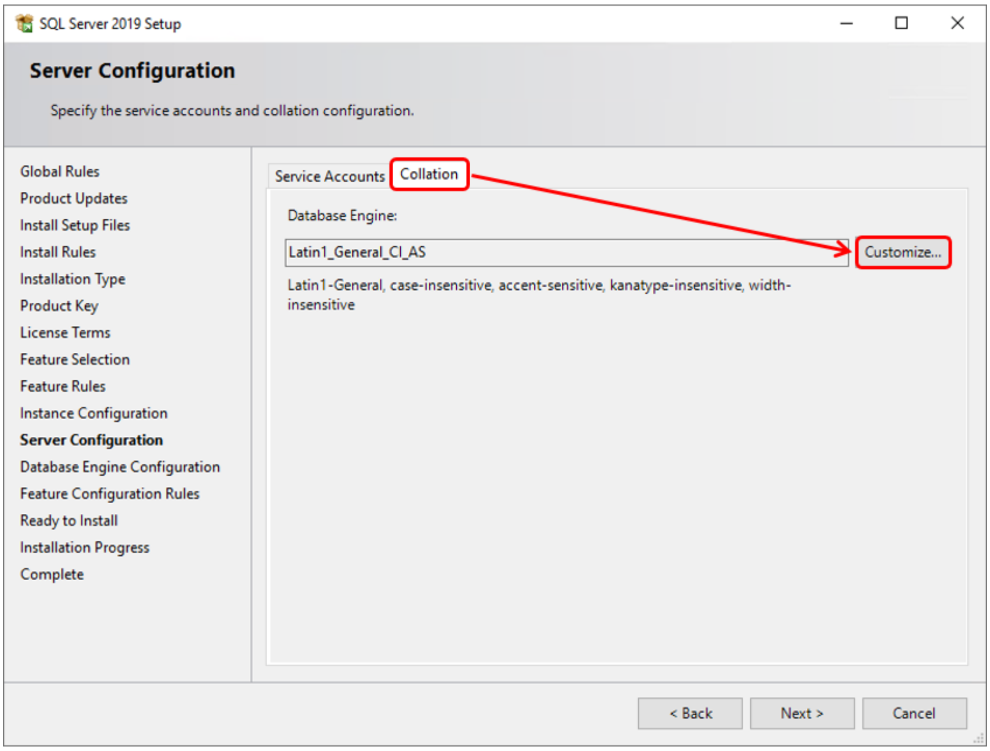


Select: **Grant Perform Volume Maintenance Task privilege to SQL Server …**

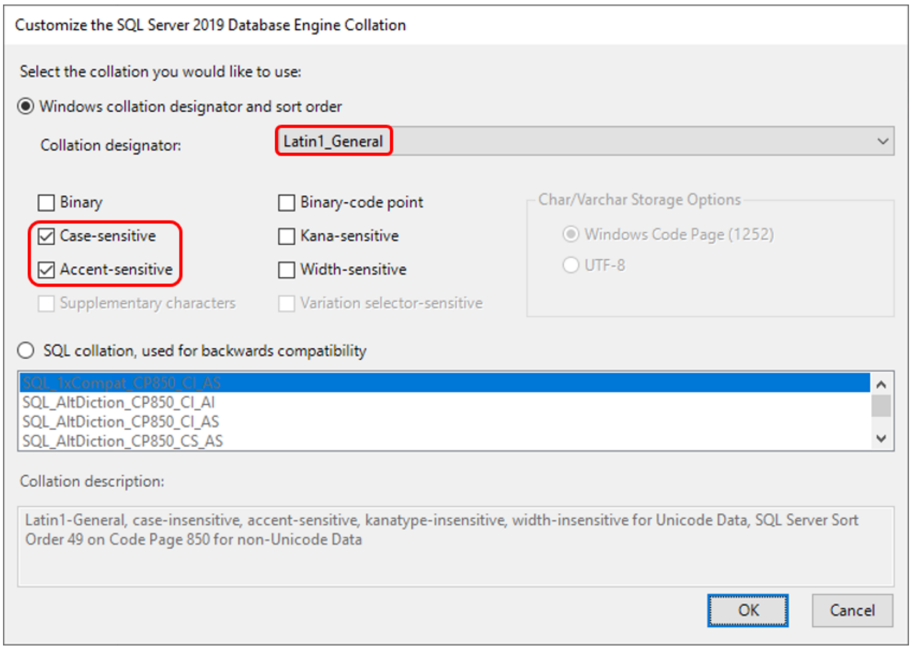
Startup Type: **Automatic** **(All services must start automatically!)**

Select Tab: **Collation**

Collation



Customize

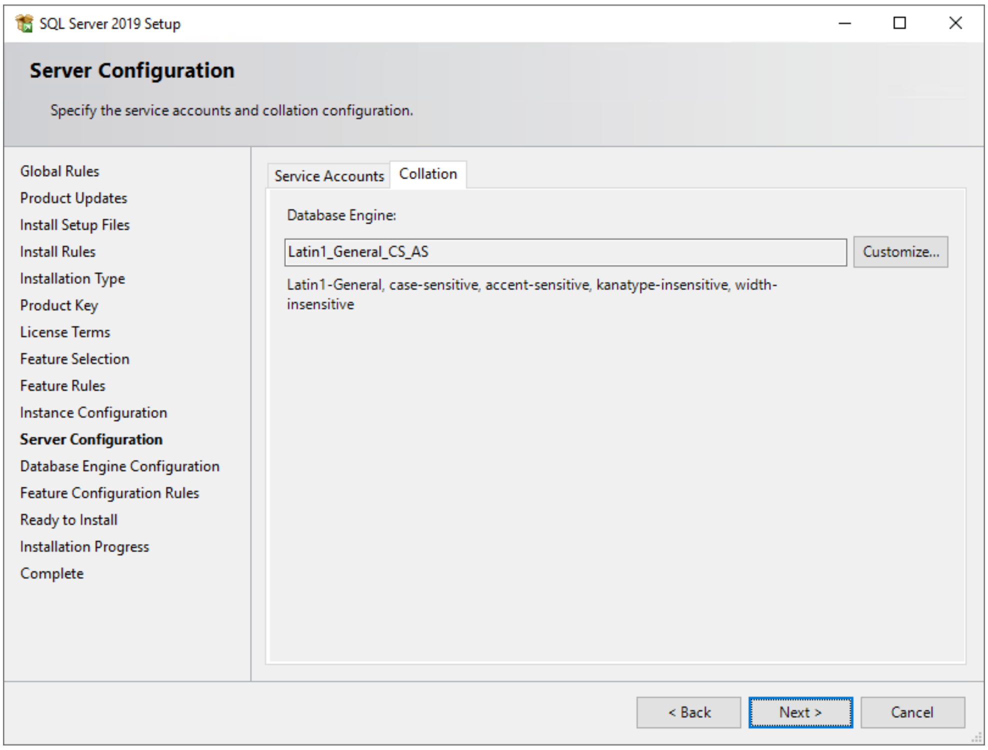


Collation designator: Latin1_General

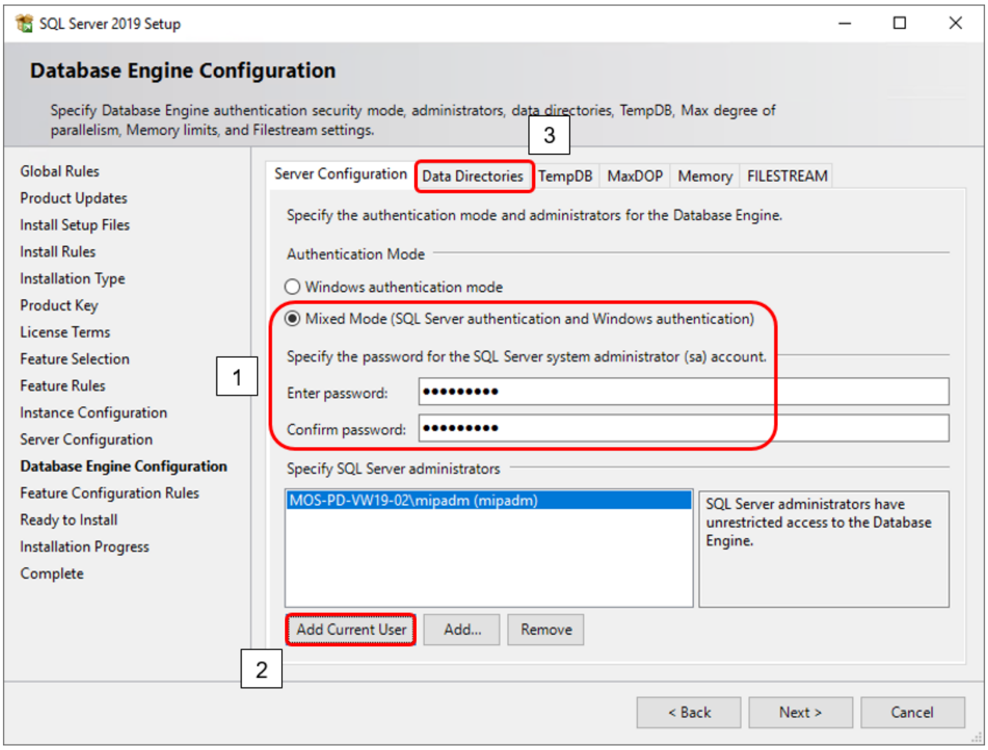
Activate: Case-sensitive

Activate: Accent-sensitive

OK



Next



Activate: **Mixed Mode** **(mandatory for MIP based products!)**

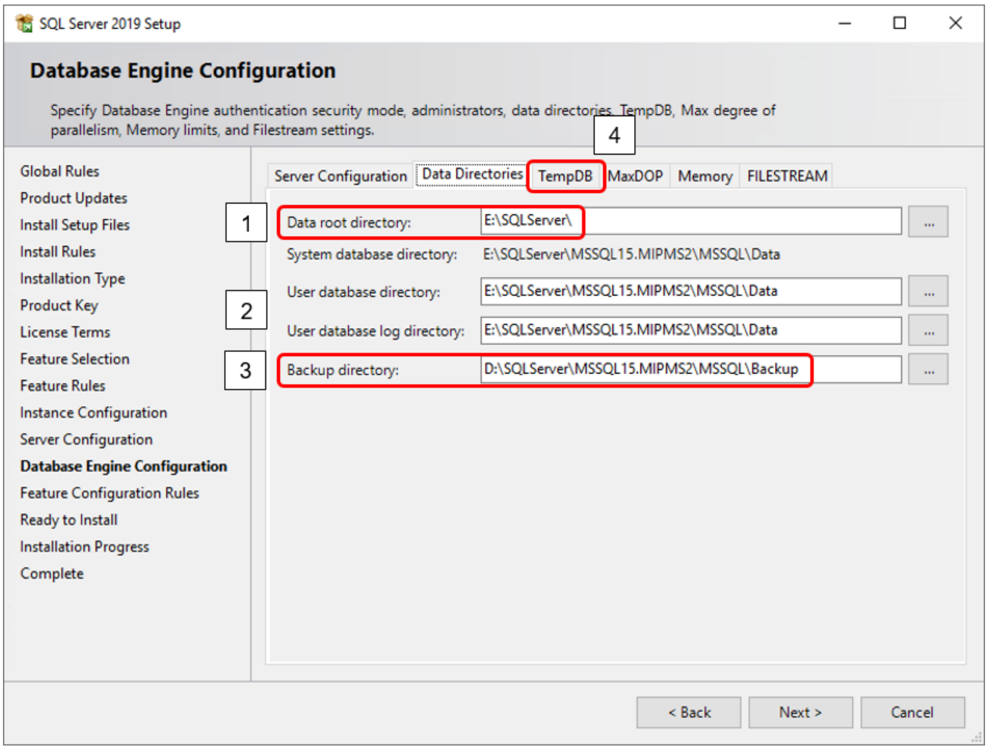
Enter password: **<SECRET sa PASSWORD>**

Confirm password: **<SECRET sa PASSWORD>** (password should be disclosed to MPDV)

Click on Button: **Add Current User**

Select Tab: **Data Directories**

Data Directories



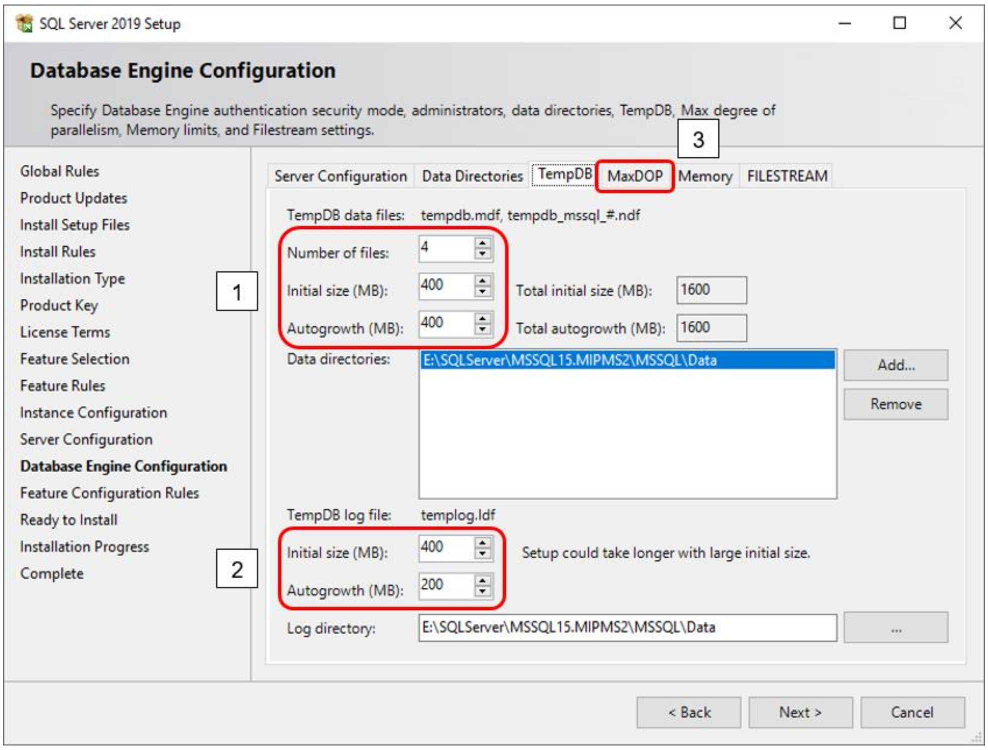
Data root directory:	**D:\SQLServer**	(when using 1 hard disk)
	E:\SQLServer	(when using 2 or more hard disks)

User database directory: **change only if necessary**
User database log dir.: **change only if necessary**

Backup directory: **D:\SQLServer\MSSQL15.MIPMS2\MSSQL\Backup**
(The backup directory should not be on the same disk as database data and log files!)

Select Tab: **TempDB**

TempDB



TempDB data files:

Number of files: 4 (Default, do not use less than 4 files!)

Initial size (MB): **400**

Autogrowth (MB): **400**

Data directories: **change only if necessary,**
e.g. to place TempDB data files on separate disks

TempDB log file:

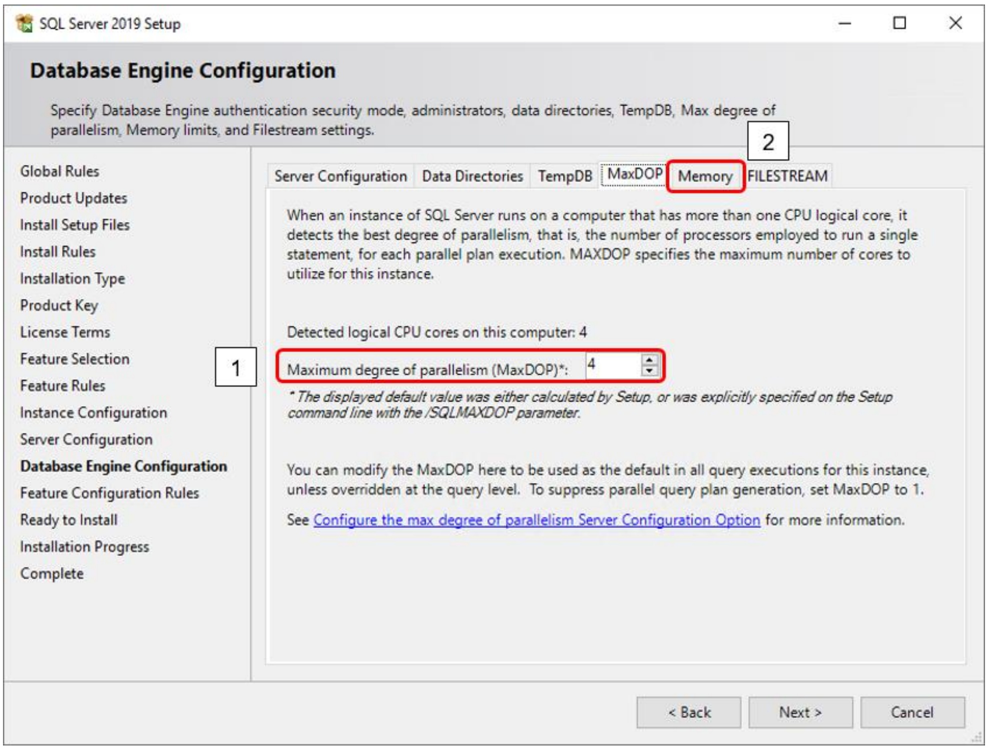
Initial size (MB): **400**

Autogrowth (MB): **200**

Log directory: **change only if necessary,**
e.g. to place TempDB log file on a separate disk

Select Tab": **MaxDOP**

MaxDOP

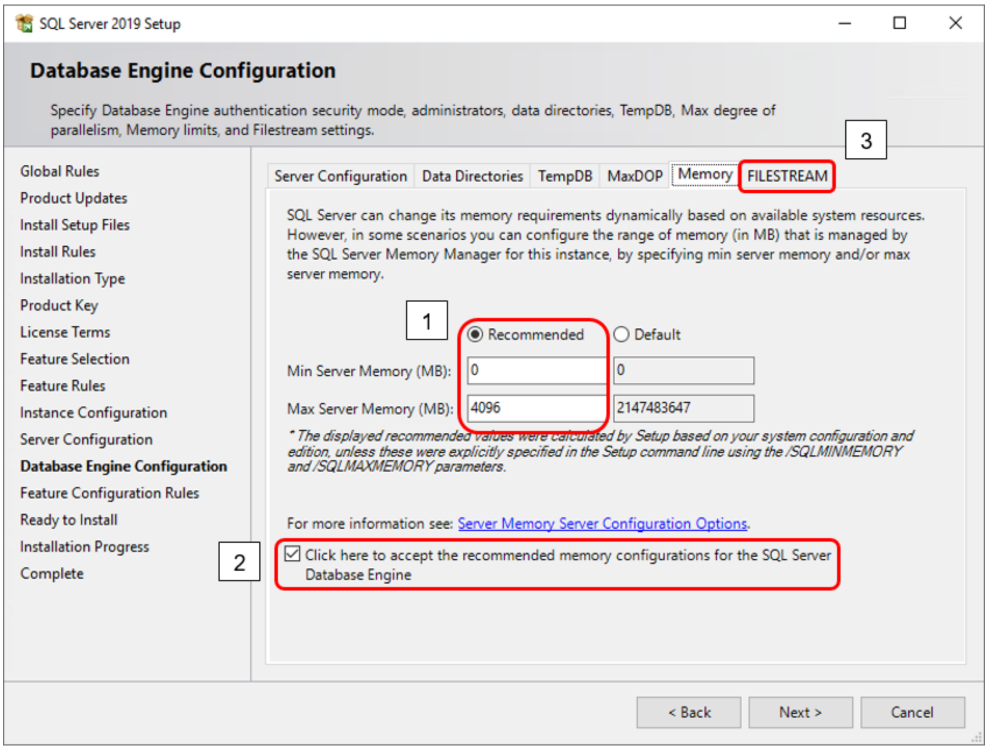


Check the setting for "**Maximum degree of parallelism (MaxDOP)**" which was calculated by Setup.

Make sure it is set to a value of at least 4

If the calculated value is higher than 4 you may leave it that way.

Select Tab: **Memory**



Change radio button from Default to Recommended:

Min Server Memory (MB): **0**

Max Server Memory (MB): **adjust according to the statement below**

(Note that the displayed recommended value does not consider the following circumstances and might need to be adjusted accordingly)

Depending on the total amount of RAM memory available on the server and the system requirements by MIP based products please set the size for “Max Server Memory (MB)” so that as much memory as possible will be used by the database instance.

The size of “Max Server Memory (MB)” will have direct influence on the performance of the MIP based application because with bigger sizes more data can be cached by the database.

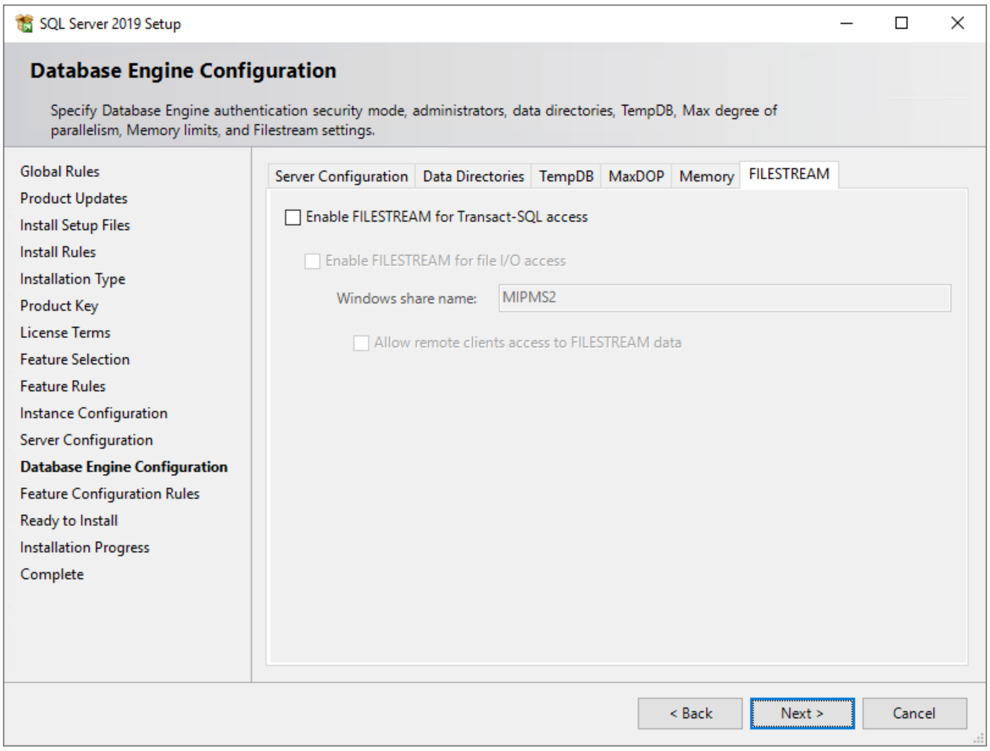
As long as it is ensured that there will be enough RAM left for the requirements of the server’s operating system (~2GB), the MIP based product services (depending on the system size ~1,5-8GB per system), the MIP WSP service (depending on the system size ~2-12GB per system) and other database instances installed on the same server you should use as much memory for “Maximum Server Memory” as possible.

Make sure that the available server memory (RAM) is used to full capacity without forcing your server into swapping.

Activate the checkbox to accept the recommended memory configurations for the SQL Server Database Engine

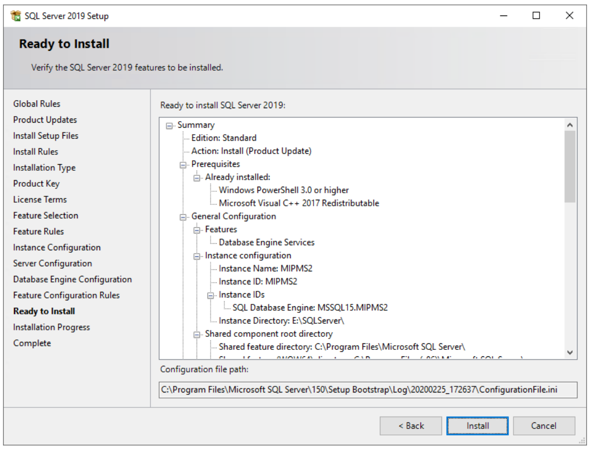
Select Tab: **FILESTREAM**

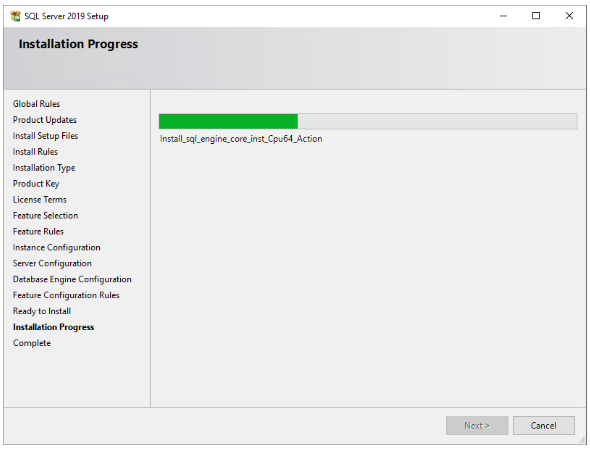
FILESTREAM



no changes necessary

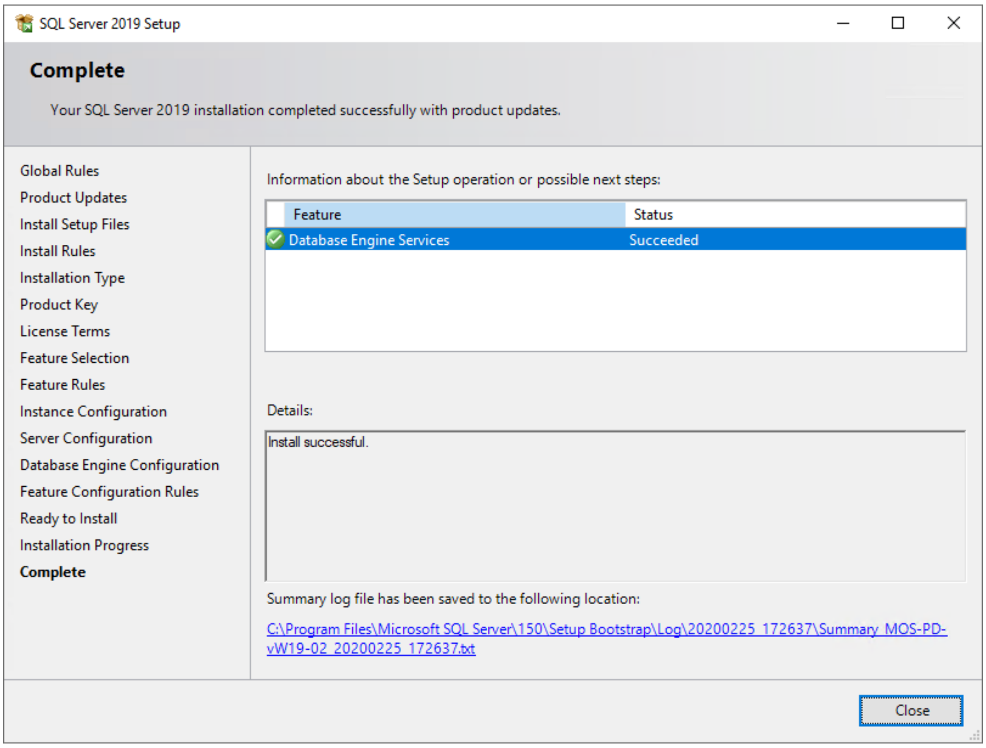
Next





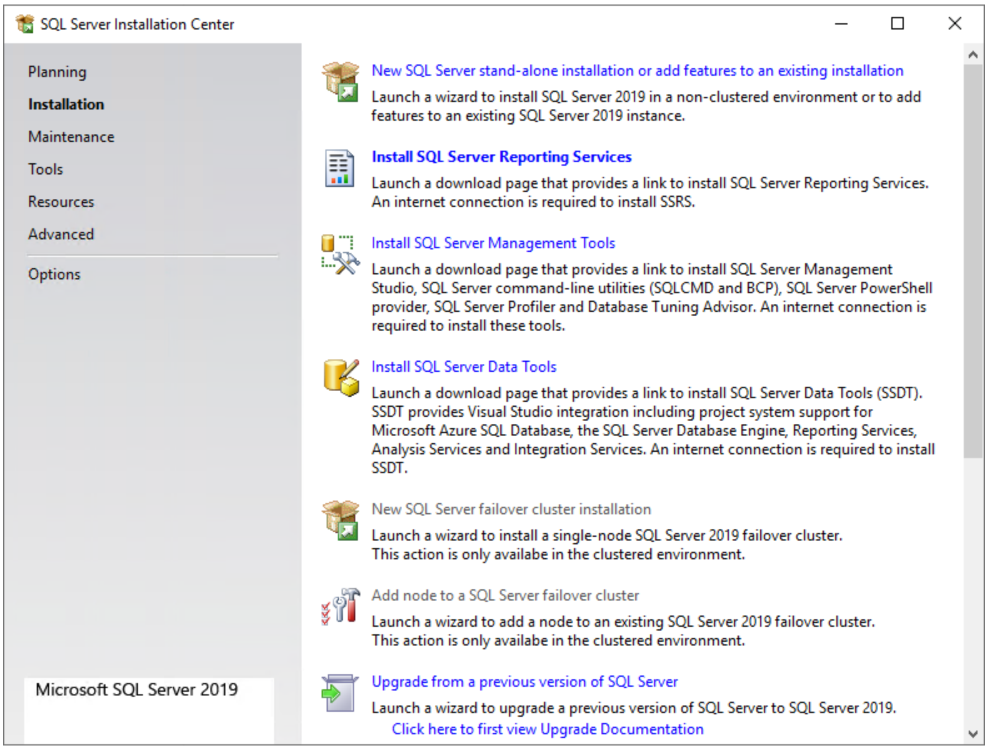
Install

Make sure the installation succeeded successfully



Close

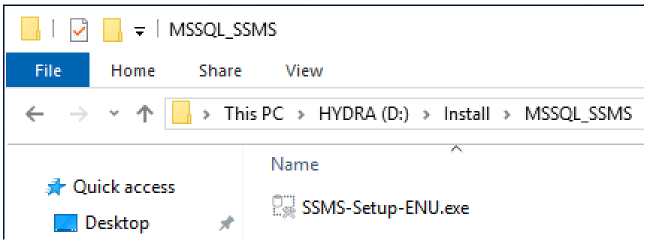
Close this Window

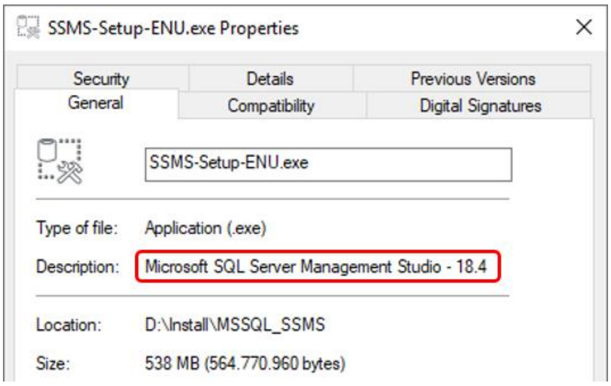


1.3. Installing SQL Server Management Studio

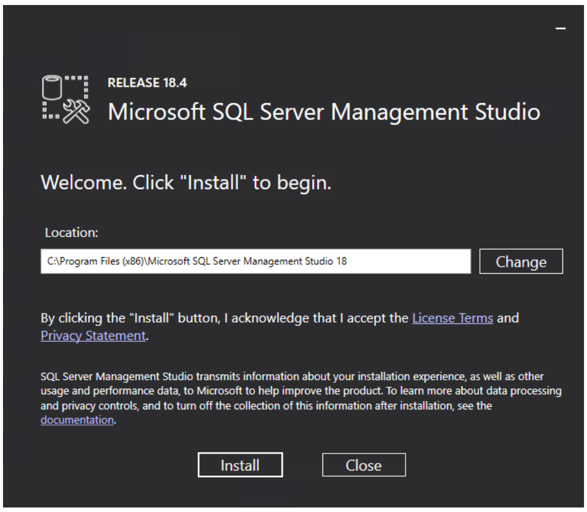
Start the Installation of the SQL Server Management Studio.
Make sure you have the installation file matching the language of your servers operating system available, e.g.:

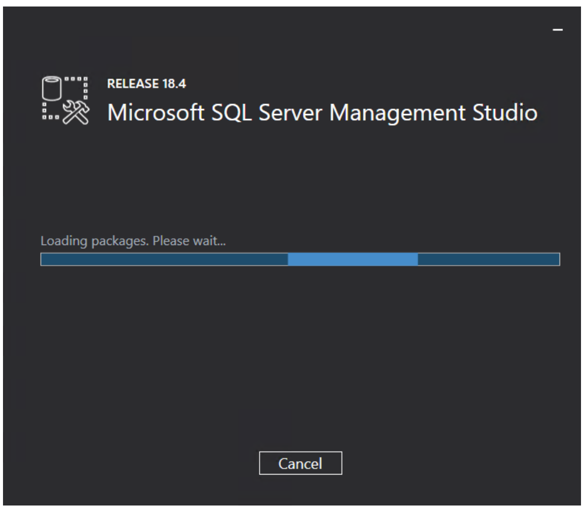
SSMS-Setup-ENU.exe



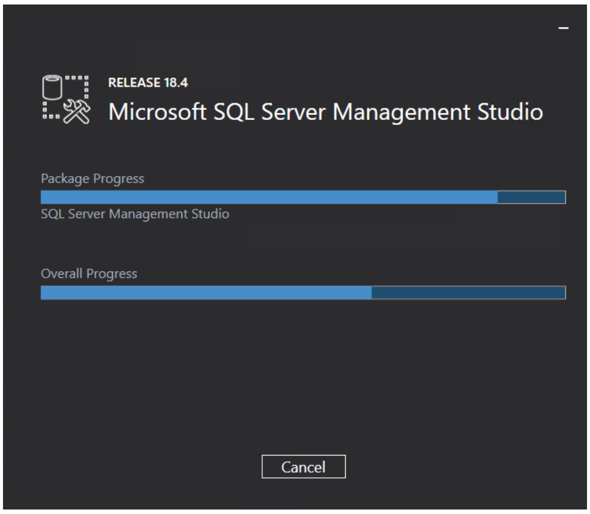


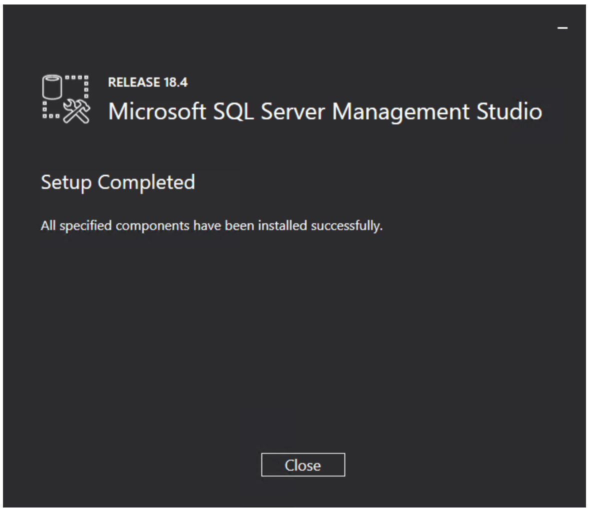
If the version of your SQL Server Management Studio is **higher than version 17** and if you are using **separate application and database servers** for your MIP based product you need to follow the instructions in chapter "1.7. Separate Database and Application Server".





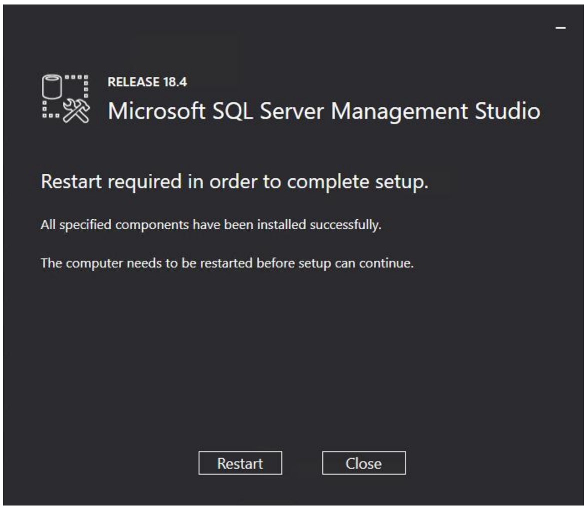
Install





Close

Depending on the installation you might see a message where a restart of the server is required.



Restart

If it is requested restart your server now.

1.4. Installing Service Pack

Currently there is no Service Pack installation mandatory for MIP based products.

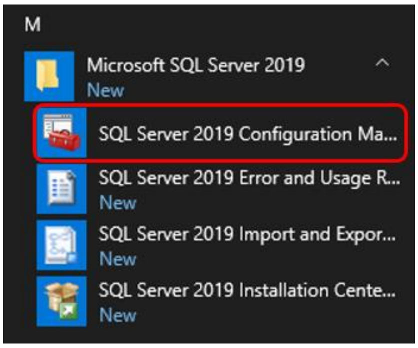
1.4.1. Installing Cumulative Update Package

Currently there is no Cumulative Update Package installation mandatory for MIP based products.

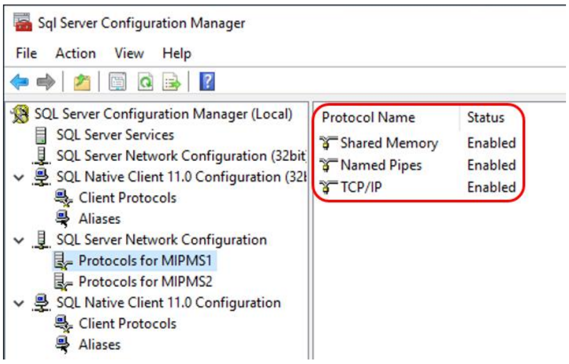
1.5. Post Installation Configuration

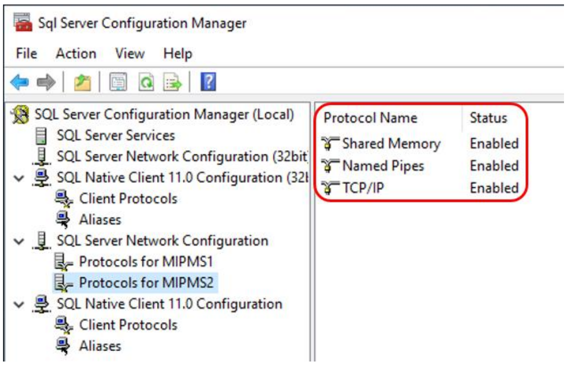
Start the “SQL Server 2019 Configuration Manager”, e.g.:

Start-Button – Microsoft SQL Server 2019 – SQL Server 2019 Configuration Manager



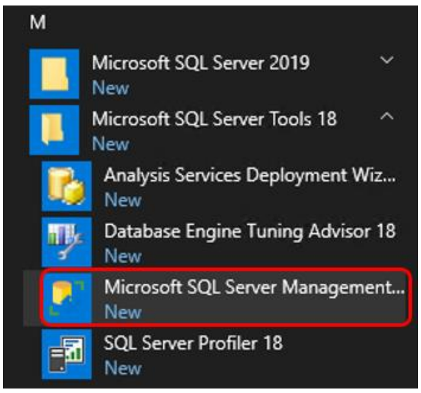
In “SQL Server Network Configuration” the protocols Shared Memory, Named Pipes and TCP/IP must be enabled for all available database instances, e.g.: MIPMS1, MIPMS2, etc.:



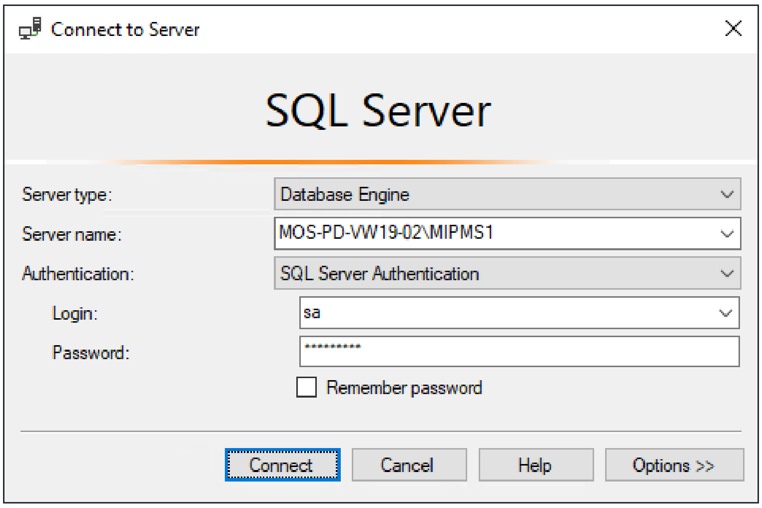


Start the “SQL Server Management Studio”, e.g.:

Start-Button – Microsoft SQL Server Tools 18 – Microsoft SQL Server Management Studio



Login to the appropriate database instance (e.g.: MIPMS1) as User "**sa**":



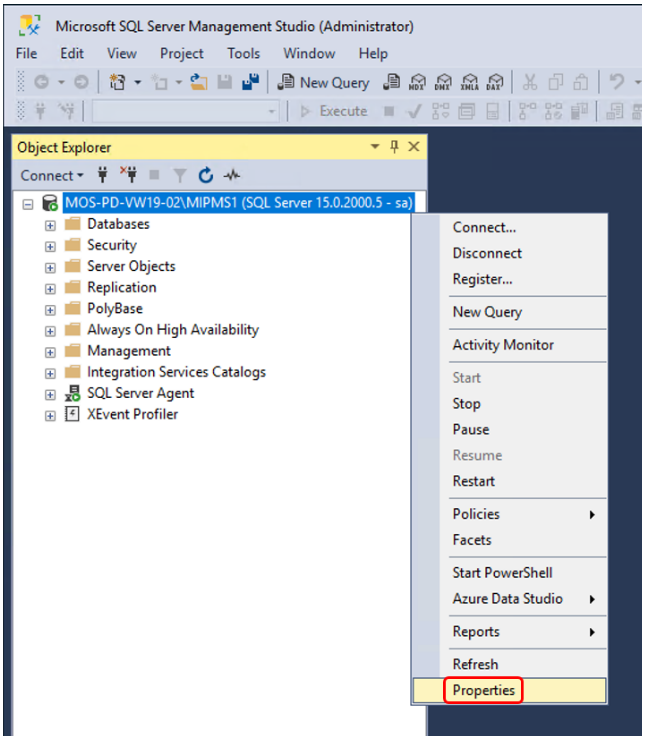
Server name: **<hostname>\MIPMS1**

Authentication: **SQL Server Authentication**

Login: **sa**

Password: **<SECRET sa PASSWORD>** (see chapter "1.2. Installing SQL Server 2019")

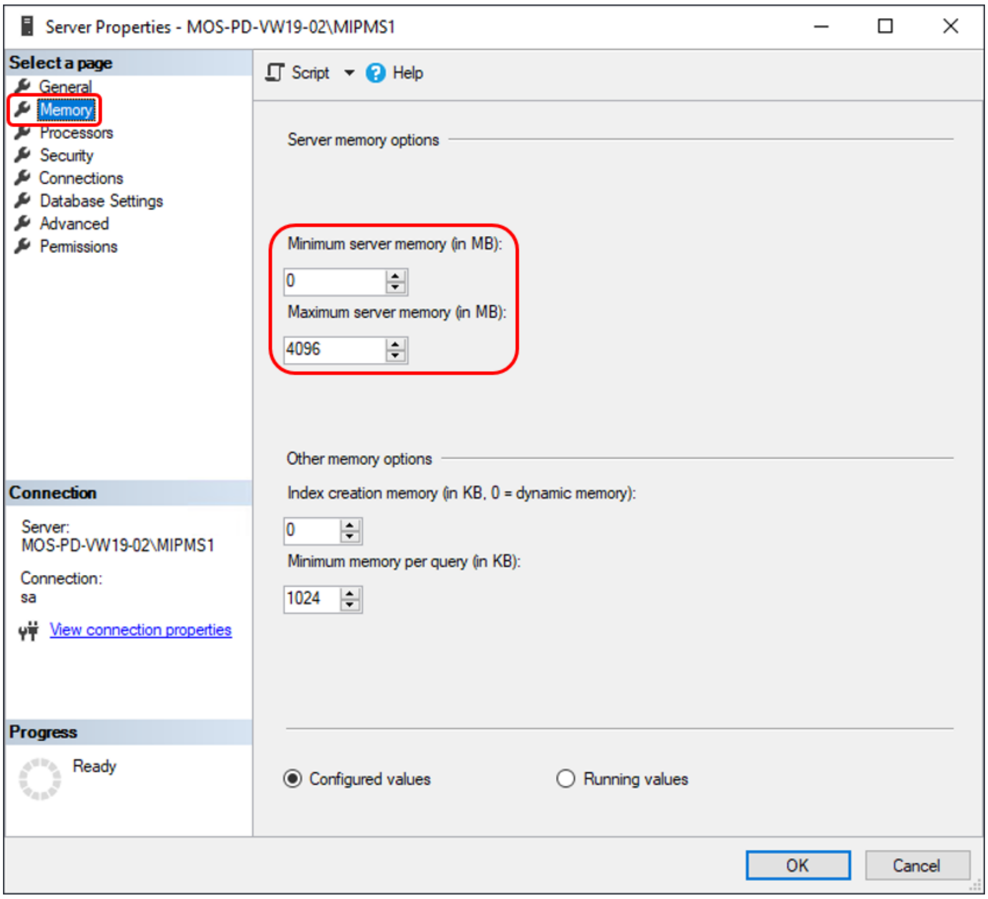
Connect



Right click on <hostname>\MIPMS1

Select: Properties

Memory (you should see the values you entered during the installation, see above)



Minimum server memory (in MB): **0**

Maximum server memory (in MB): **as much as available**

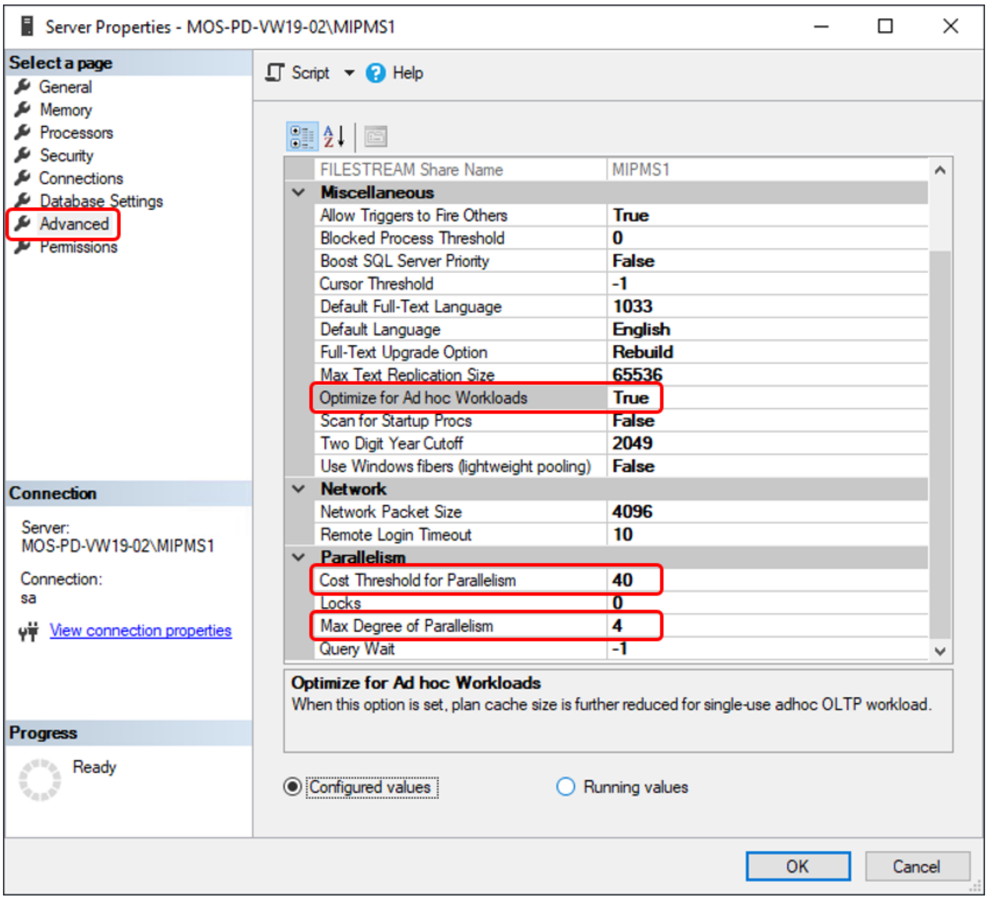
Depending on the total amount of RAM memory available on the server and the system requirements by MIP based products please set the size for "Maximum server memory" so that as much memory as possible will be used by the database instance.

The size of "Maximum server memory" will have direct influence on the performance of the MIP based application because with bigger sizes more data can be cached by the database.

As long as it is ensured that there will be enough RAM left for the requirements of the server's operating system (~2GB), the MIP based product services (depending on the system size ~1,5-8GB per system), the MIP WSP service (depending on the system size ~2-12GB per system) and other database instances installed on the same server you should use as much memory for "Maximum server memory" as possible.

Make sure that the available server memory (RAM) is used to full capacity without forcing your server into swapping.

Advanced



Optimize for Ad hoc Workloads	**True**
Cost Threshold for Parallelism	**40**
Max Degree of Parallelism	**4** or higher (see MaxDOP during the installation above)
OK	

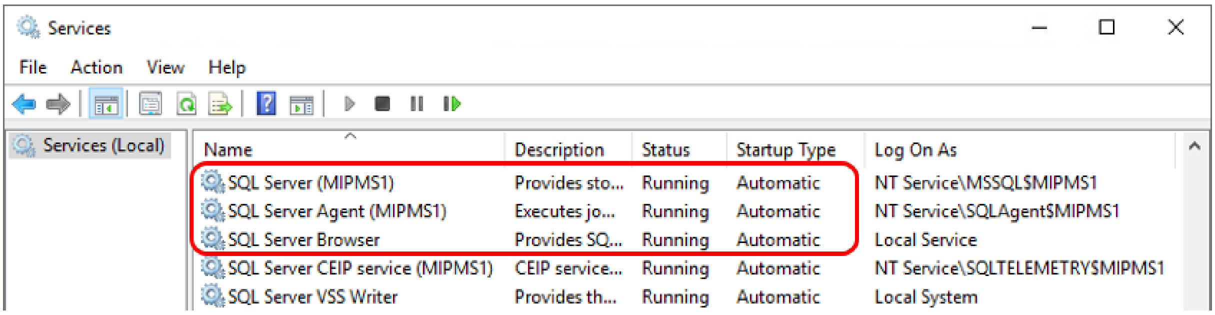
Repeat those configurations for every database instance.

Start-Button – Windows Administrative Tools – Services

The following 3 SQL Server services are required by MIP based products.

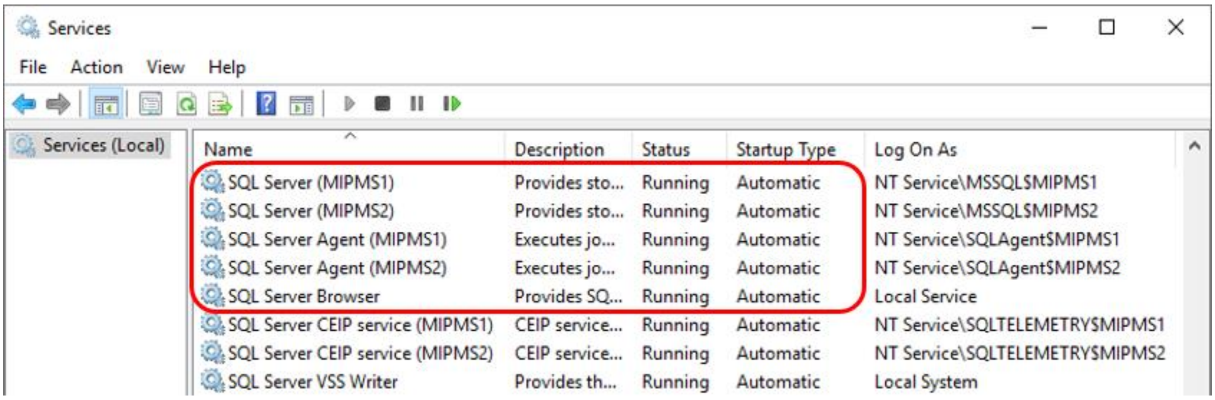
A running service "**SQL Server Browser**" is mandatory for the functionality of MIP based products.

Check that those services are starting automatically:



Name	Description	Status	Startup Type	Log On As
SQL Server (MIPMS1)	Provides sto...	Running	Automatic	NT Service\MSSQL$MIPMS1
SQL Server Agent (MIPMS1)	Executes jo...	Running	Automatic	NT Service\SQLAgent$MIPMS1
SQL Server Browser	Provides SQ...	Running	Automatic	Local Service
SQL Server CEIP service (MIPMS1)	CEIP service...	Running	Automatic	NT Service\SQLTELEMETRY$MIPMS1
SQL Server VSS Writer	Provides th...	Running	Automatic	Local System

Example for a multi system installation:



Name	Description	Status	Startup Type	Log On As
SQL Server (MIPMS1)	Provides sto...	Running	Automatic	NT Service\MSSQL$MIPMS1
SQL Server (MIPMS2)	Provides sto...	Running	Automatic	NT Service\MSSQL$MIPMS2
SQL Server Agent (MIPMS1)	Executes jo...	Running	Automatic	NT Service\SQLAgent$MIPMS1
SQL Server Agent (MIPMS2)	Executes jo...	Running	Automatic	NT Service\SQLAgent$MIPMS2
SQL Server Browser	Provides SQ...	Running	Automatic	Local Service
SQL Server CEIP service (MIPMS1)	CEIP service...	Running	Automatic	NT Service\SQLTELEMETRY$MIPMS1
SQL Server CEIP service (MIPMS2)	CEIP service...	Running	Automatic	NT Service\SQLTELEMETRY$MIPMS2
SQL Server VSS Writer	Provides th...	Running	Automatic	Local System

Restart your server now.

1.6. Database

Other than MIP the applications HYDRA and FEDRA allow for a multi system installation (multiple HYDRA or FEDRA systems on the same (application) server).
For multi system installations every HYDRA or FEDRA system needs its own database.
Name your databases **mip1**, **mip2**, **mip3**, etc.
With multi system installations it is strongly recommended that every database has its own database instance available, e.g.: **MIPMS1**, **MIPMS2**, **MIPMS3**, etc.

Before you can proceed with creating a database to be used by the MIP based product you need to have the following files available:

db_mip_sql2019.sql (or db_mip_sql2019.**bsp**)
create_user_mipadm_sql201x.sql

Either those files were provided to you in advance by MPDV or you might have installation packages for the MIP based application available.
In the MIP package the files are located in the ".\Database" directory.
In the HYDRA or the FEDRA package the files are located in the ".\db_sql" directory.

1.6.1. Create Database

Attention:
Do not run these installation procedures on a computer where MIP based products are already running.
By doing so, you might destroy your existing MIP based system.
Please be extremely cautious if you decide to do it anyway.

Logon to the (database) server as local user "**mipadm**".

Check if the database creation script **db_mip_sql2019.sql** is available in the above mentioned directories.
If not, make a copy of the file **db_mip_sql2019.bsp**, e.g.:
copy d:\mip1\db_sql\db_mip_sql2019.bsp d:\mip1\db_sql\db_mip_sql2019.sql

Edit the database creation script and make sure that all path configurations are set corresponding to your system environment, e.g.:
FILENAME=N'**E:\SQLServer\MSSQL15.MIPMS1\MSSQL\DATA**rootdbs_Data.MDF'
All paths used in the database creation script must exist on your server before running the script!

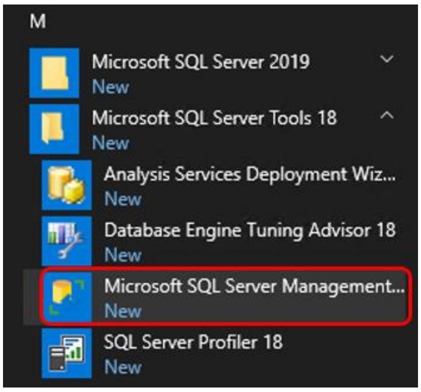
Do not change or remove settings regarding user configurations and authorizations or any other permissions without consulting with MPDV first!
By doing so it might be possible that the MIP based product will not work properly.

Create a backup copy of the edited database creation script **db_mip_sql2019.sql**, e.g.:

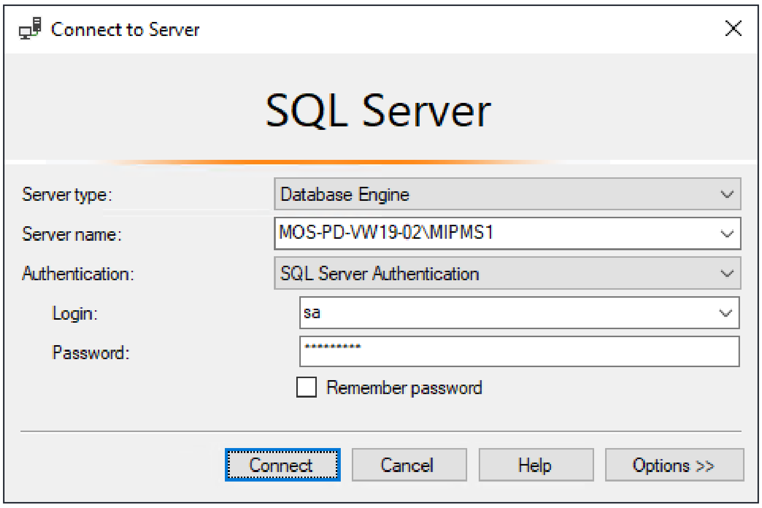
`copy d:\mip1\db_sql\db_mip_sql2019.sql d:\mip1\db_sql\db_mip_sql2019_mip1.sql`

Start the “SQL Server Management Studio”, e.g.:

Start-Button – Microsoft SQL Server Tools 18 – Microsoft SQL Server Management Studio



Login to the appropriate database instance (e.g.: MIPMS1) as User “**sa**”:



Server name: **<hostname>\MIPMS1**

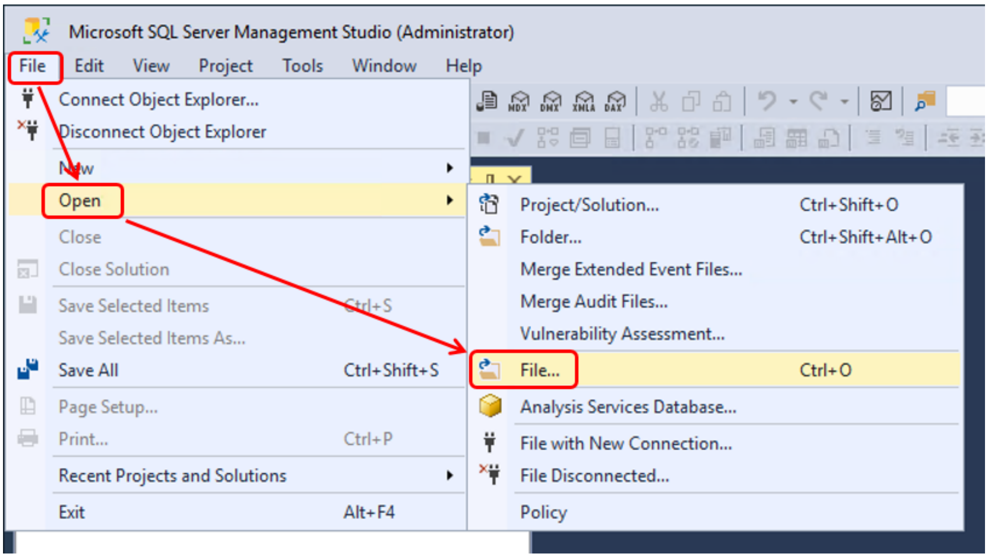
Authentication: **SQL Server Authentication**

Login: **sa**

Password: **<SECRET sa PASSWORD>** (see chapter “1.2. Installing SQL Server 2019”)

Connect

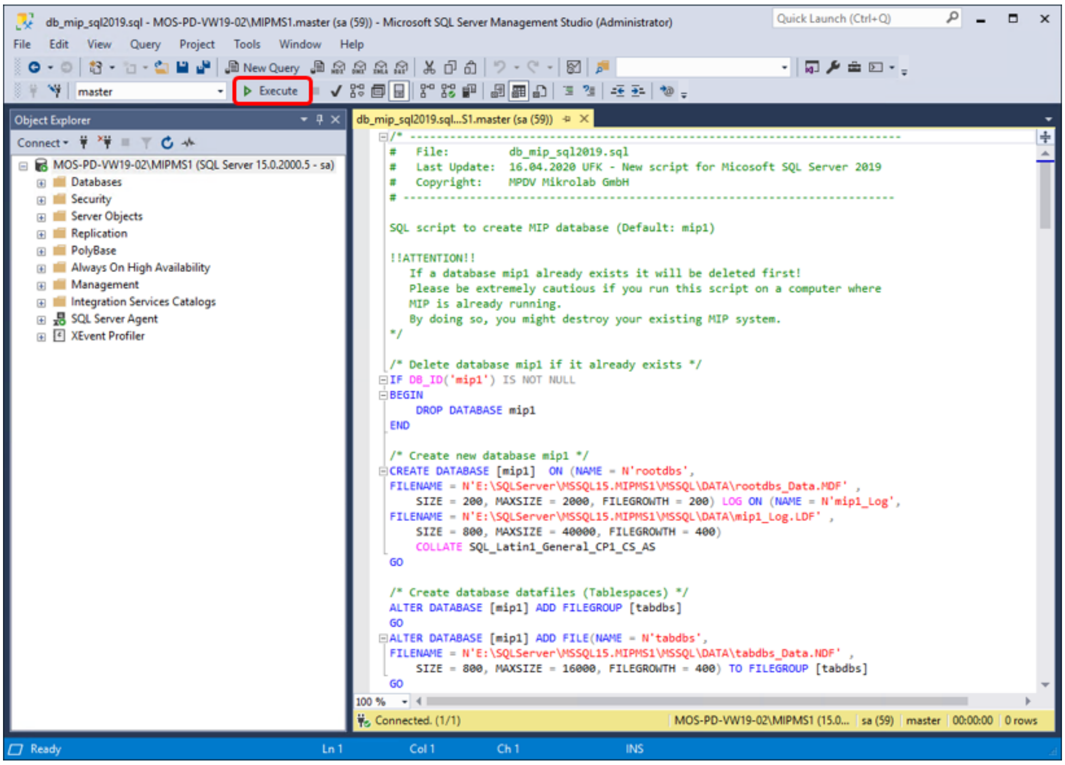
Open the database creation script (e.g. db_mip_sql2019.sql) in menu **"File → Open → File…"**



Select file, e.g.: **d:\mip1\db_sql\db_mip_sql2019.sql**

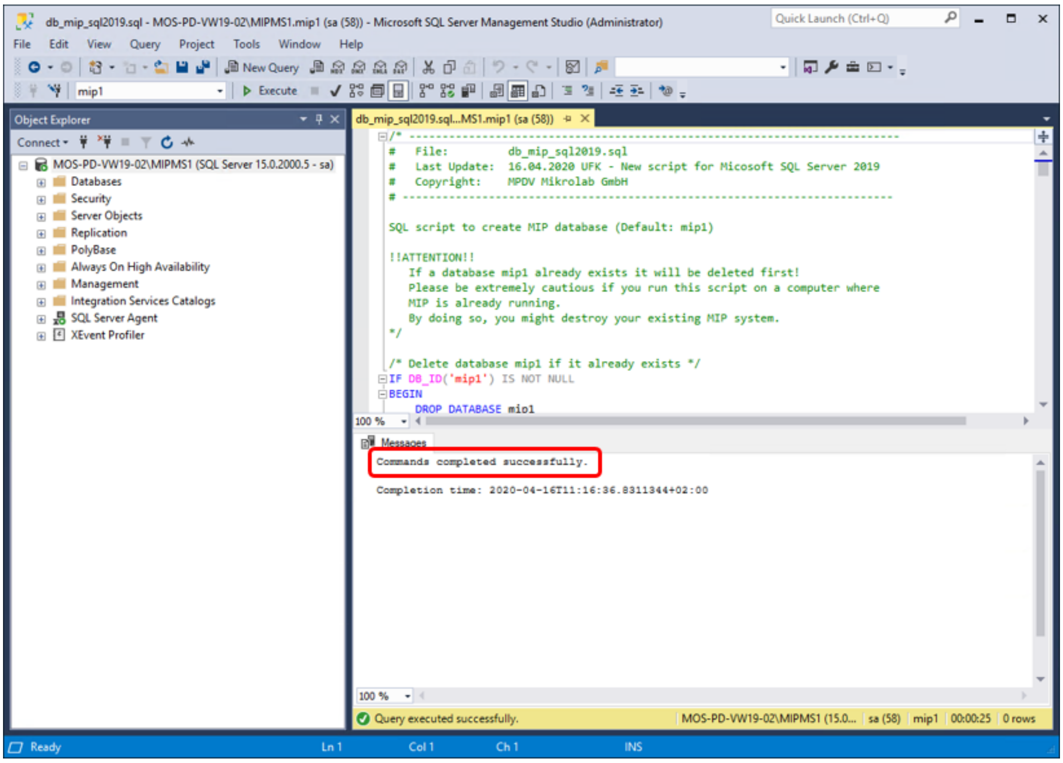
Attention:

The database creation script will delete an existing database (Default: "mip1") before creating a new one. If you perform the following procedures on a server where there are already MIP based databases installed you might destroy your existing MIP based system.



Execute

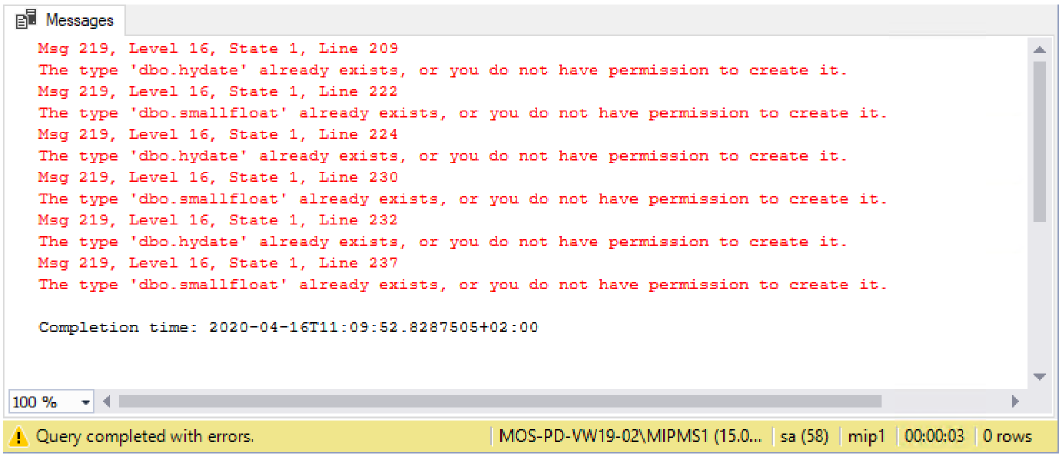
The database creation must complete without errors:



→ Commands completed successfully

If a MIP based database was already created earlier on this server there will be error messages about user defined data types “dbo.hydate” and “dbo.smallfloat” which already exist.

In this case the following messages can be ignored:



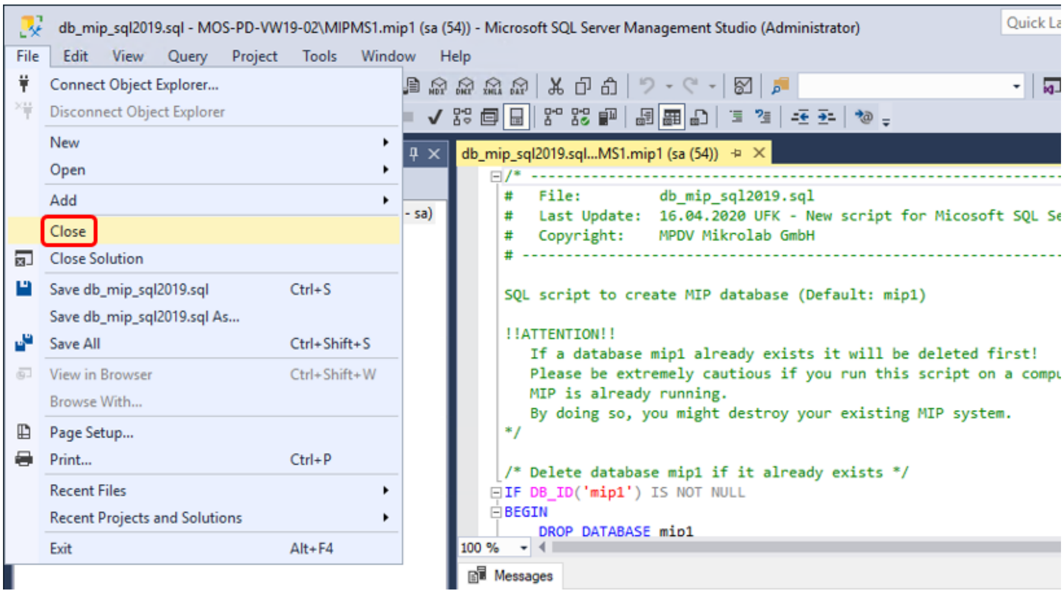
To avoid those error messages you might want to delete the user defined data types first before running the database creation script again.
In this case use the following commands:

```
USE model
DROP TYPE hydate;
DROP TYPE smallfloat;

USE tempdb
DROP TYPE hydate;
DROP TYPE smallfloat;

USE mip1
DROP TYPE hydate;
DROP TYPE smallfloat;
```

Close session:



File → Close

Repeat these procedures for every additional database instance (e.g.: MIPMS2) and its database (e.g.: mip2).

The database creation script “db_mip_sql2019.sql” needs to be duplicated and adjusted accordingly.
Make sure to save a copy of the new script using a unique file name, e.g.:
d:\mip1\db_sql\db_mip_sql2019_mip2.sql

1.6.2. Create Database User

Note about database user name "mipadm":

MPDV strongly recommends not to change the default database user name "mipadm"!

If you decide to change it anyway it is mandatory that the names of the database user and the database schema **are always identical!**

If the database user name needs to be different than "mipadm" the schema name has to be changed accordingly.

At present only characters "**a-z**" and "**0-9**" are allowed!

Upper case characters are **not** allowed!

Note about password for database user "mipadm":

It is possible to use different passwords than MPDVs default password.

For security reasons MPDV strongly recommends to change the default password used for the database user "mipadm" as soon as the user is successfully created (see below).

The new password must be disclosed to MPDV.

The following characters are **not** allowed for the password:

- Characters with ASCII Code **> 126** (e.g. German umlauts, French "umlauts", etc.)
- Pipe: **|**
- Ampersand: **&**

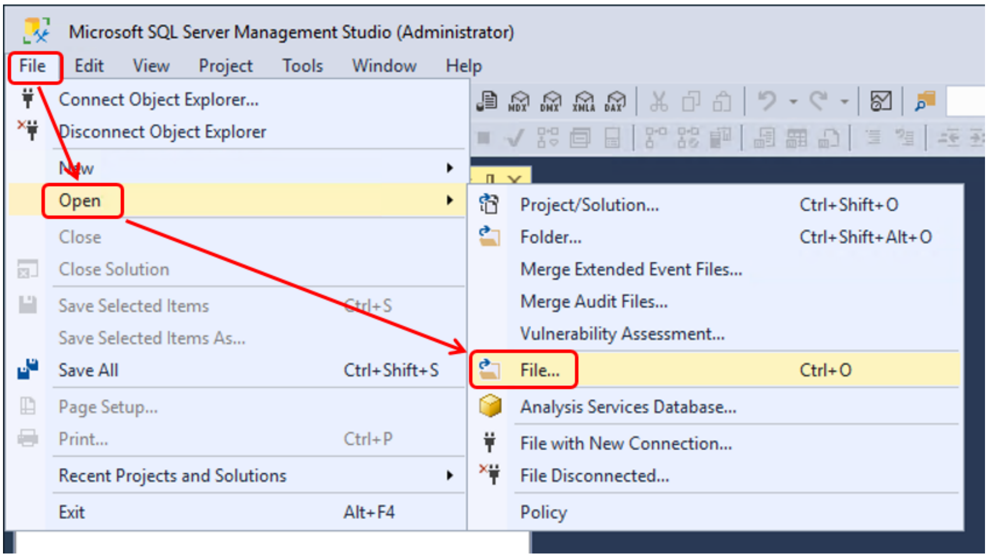
Do not change or remove settings regarding user configurations and authorizations or any other permissions without consulting with MPDV first!

By doing so it might be possible that the MIP based product will not work properly.

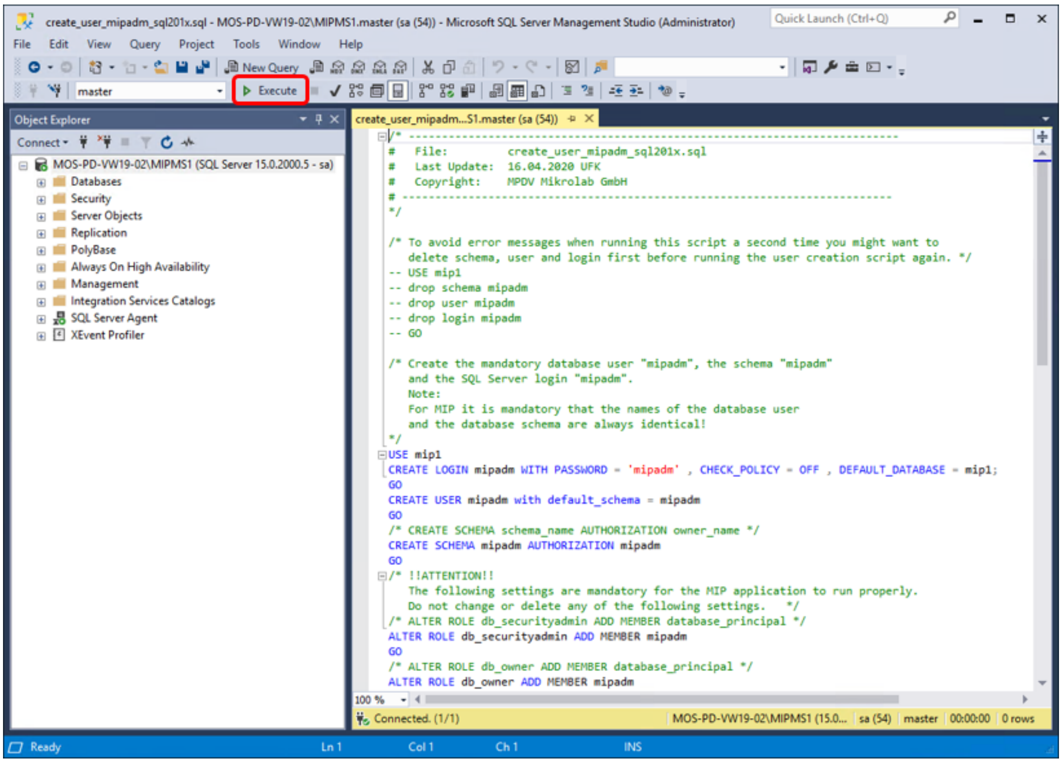
Make sure that no other sessions are open before running the script.

Open database user creation script (create_user_mipadm_sql201x.sql) in menu

“File → Open → File…”



Select file, e.g.: **d:\mip1\db_sql\create_user_mipadm_sql201x.sql**

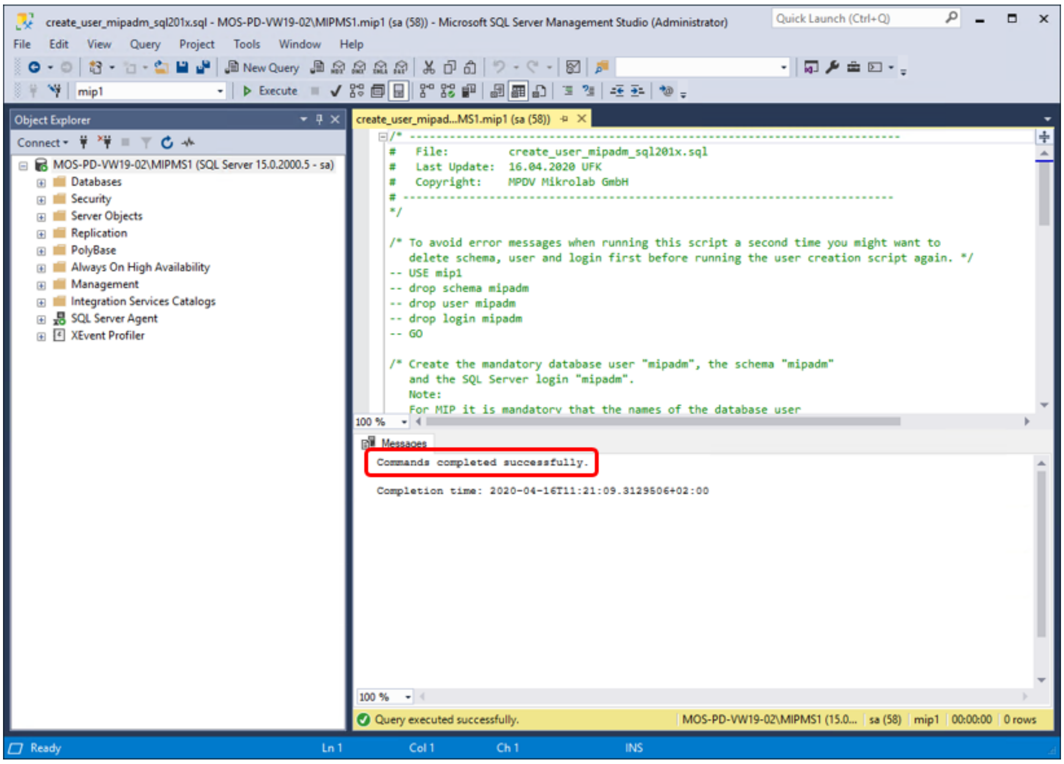


For security reasons MPDV strongly recommends to change the password for the database user “mipadm” as soon as the user is successfully created (see below).

If you decide to change the password directly here in the SSMS window make sure not to save the user creation script afterwards which then will contain the supposedly secret password in plain text!

Execute

The user must be created without errors:



→ **Commands completed successfully**

For security reasons MPDV strongly recommends to change the password for the database user “mipadm” as soon as the user is successfully created (see above).

To change a user’s password use the following commands:

```
USE master
ALTER LOGIN login_name WITH PASSWORD = 'password'
```

e.g.:

```
USE master
ALTER LOGIN mipadm WITH PASSWORD = 'NewSecurePW'
```

The new password must be disclosed to MPDV.

If the user “mipadm” was already created earlier on this server the following messages can be ignored when executing the user creation script again:

```
Messages
Msg 15025, Level 16, State 1, Line 23
The server principal 'mipadm' already exists.
Msg 15023, Level 16, State 1, Line 25
User, group, or role 'mipadm' already exists in the current database.
Msg 2714, Level 16, State 6, Line 28
There is already an object named 'mipadm' in the database.
Msg 2759, Level 16, State 0, Line 28
CREATE SCHEMA failed due to previous errors.

Completion time: 2020-04-16T10:47:12.2407001+02:00
```

To avoid those error messages you might want to delete schema, user and login first before running the user creation script again.

In this case use the following commands:

```
USE mip1
drop schema mipadm
drop user mipadm
drop login mipadm
GO
```

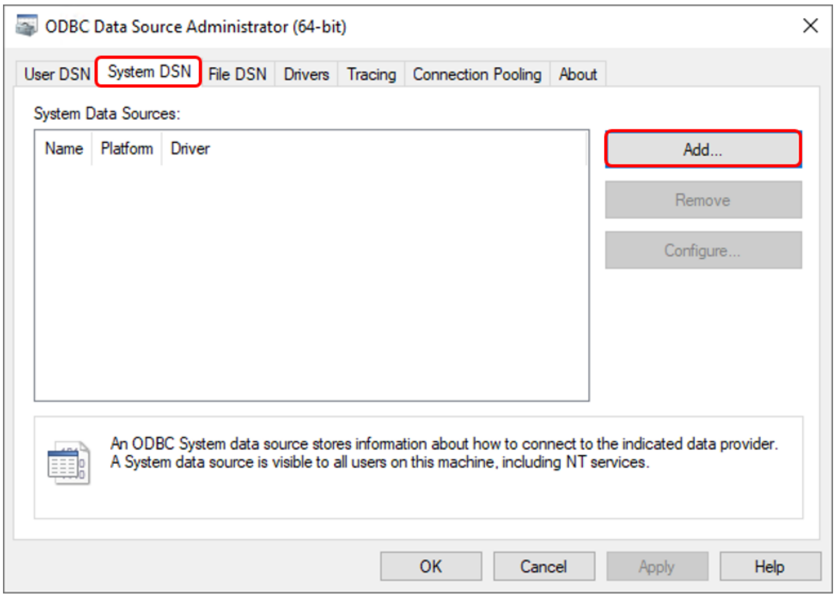
Repeat these procedures for every additional database instance (e.g.: MIPMS2) and its database (e.g.: mip2).

The user creation script “create_user_mipadm_sql201x.sql” needs to be duplicated and adjusted accordingly. Make sure to save a copy of the new script using a unique file name, e.g.:

d:\mip1\db_sql\create_user_mipadm_sql201x_mip2.sql

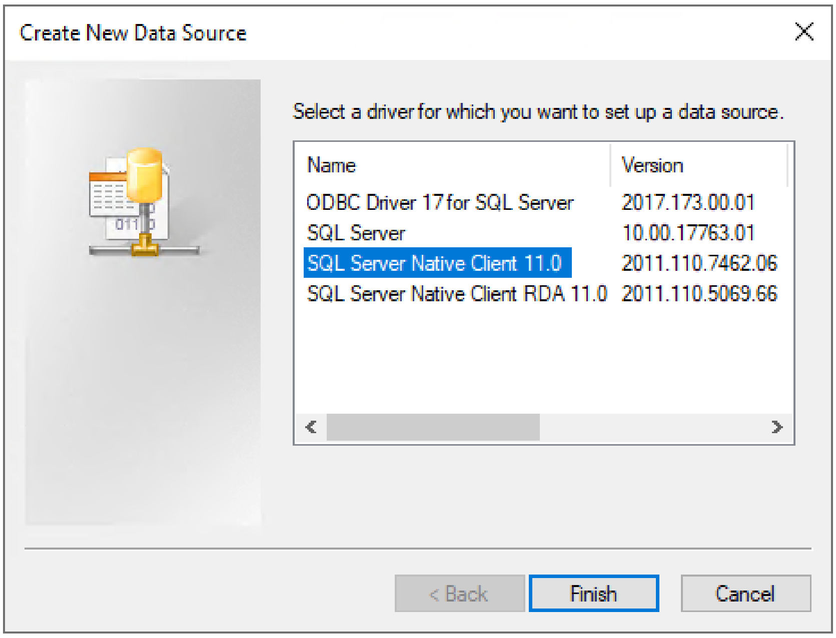
1.6.3. ODBC Configuration

Start-Button – Windows Administrative Tools – ODBC Data Sources (64-bit)



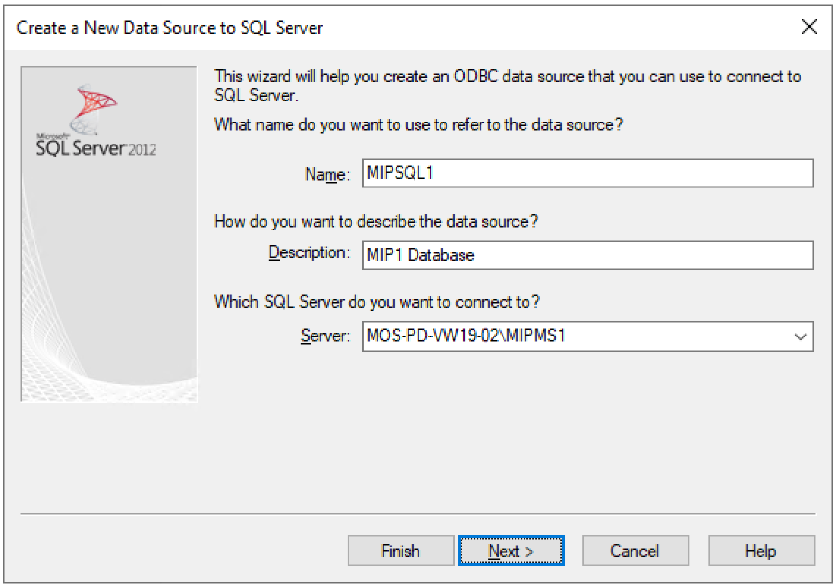
Select: System DSN

Select: Add



Select: SQL Server Native Client 11.0

Finish



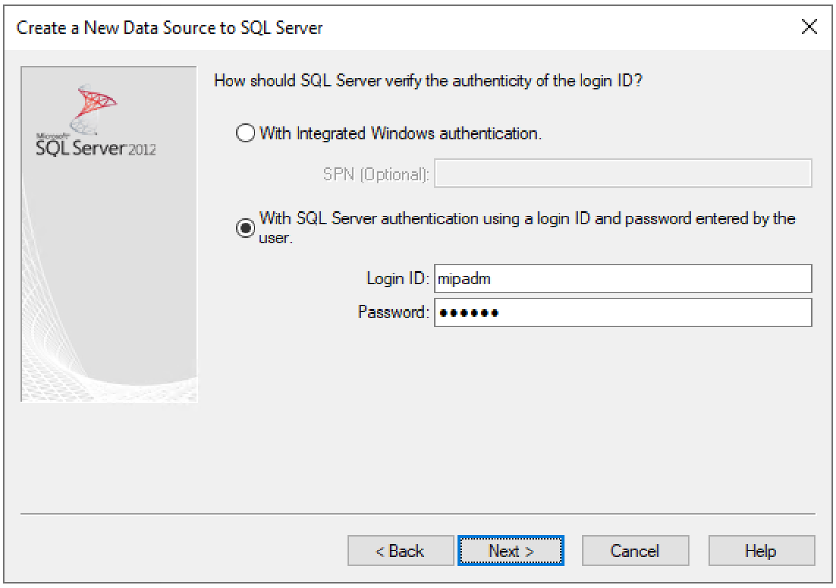
Name: **MIPSQL1** (mandatory for MIP based products, e.g. for HYDRA system 1)
(use MIPSQL2, MIPSQL3, etc. for additional HYDRA systems)

Description: **MIP1 Database**

Server: **<hostname>\MIPMS1**

(<hostname> = hostname or IP-address of the MIP based products (database) server)

Next

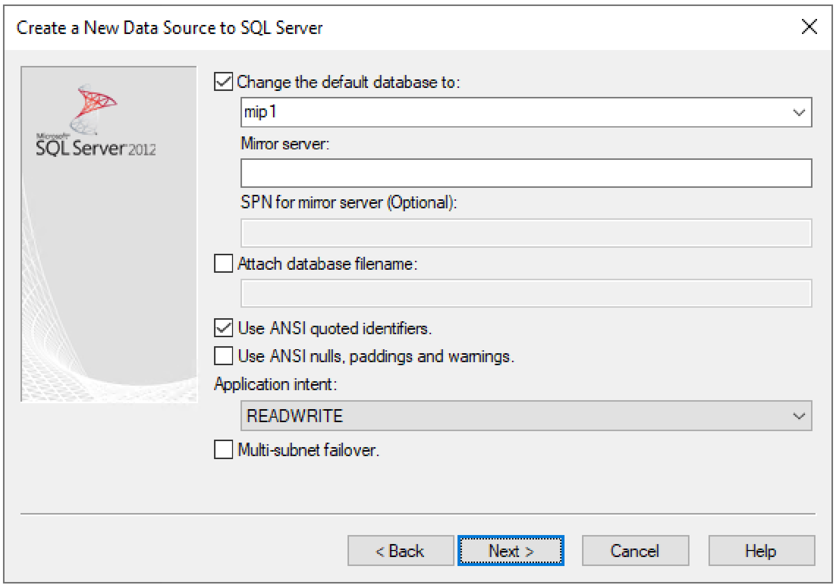


Select: **With SQL Server authentication using a login ID and password entered by the user**

Login ID: **mipadm** (see chapter “1.6.2 Create Database User”)

Password: **<SECRET mipadm PASSWORD>** (see chapter “1.6.2 Create Database User”)

Next



Change the default database to: **mip1**

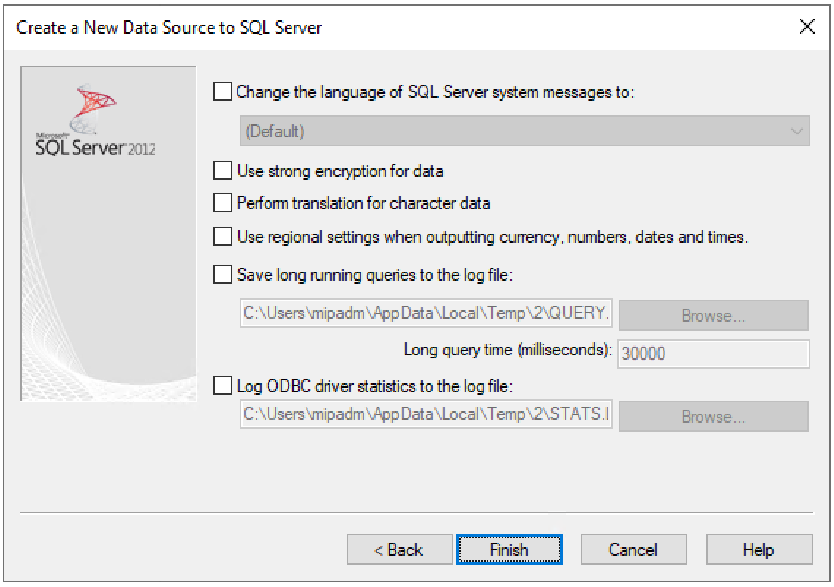
(select mip2, mip3, etc. for additional HYDRA systems)

Activate: Use ANSI quoted identifiers

Deactivate: Use ANSI nulls, paddings and warnings

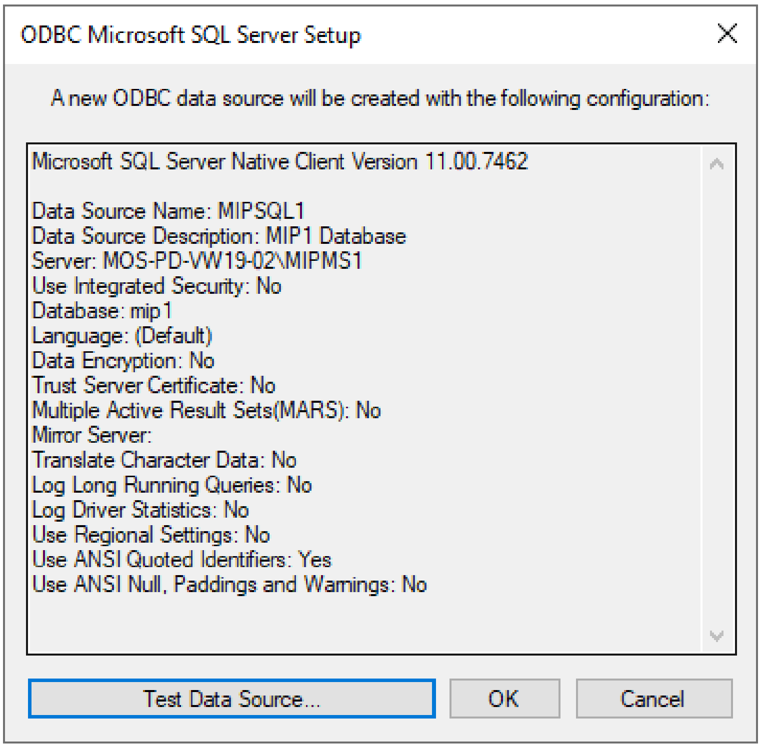
Application intent: READWRITE

Next



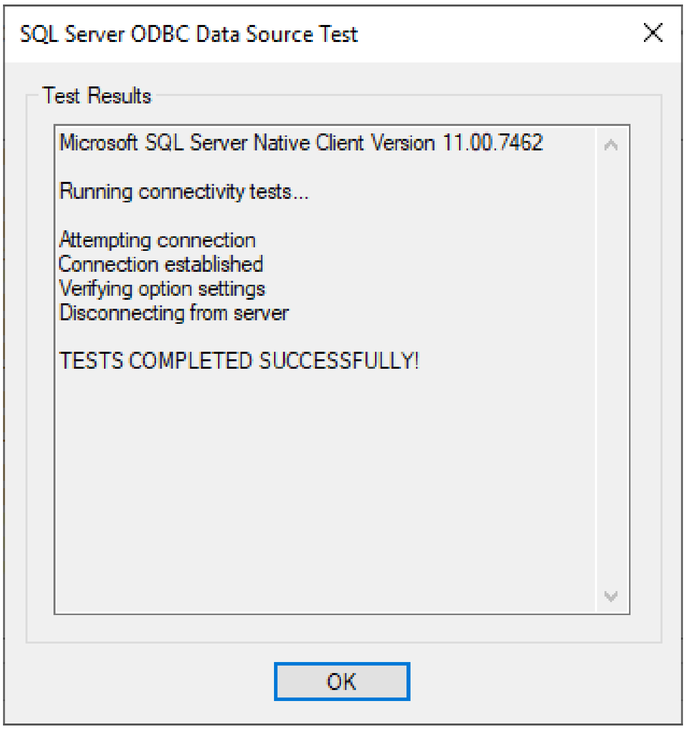
Deactivate all!

Finish

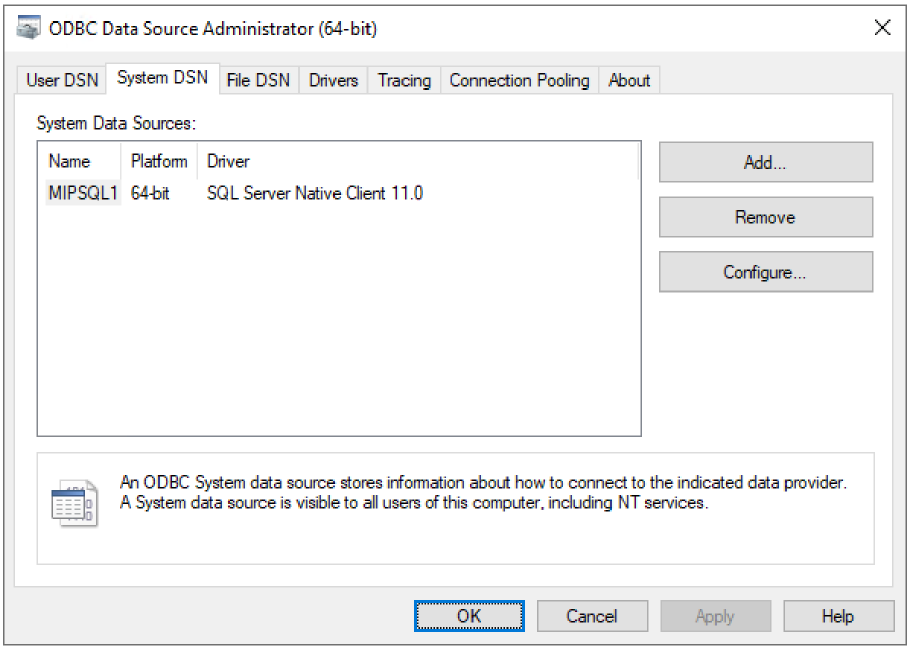


Test Data Source

Make sure the connectivity tests completed successfully:



OK

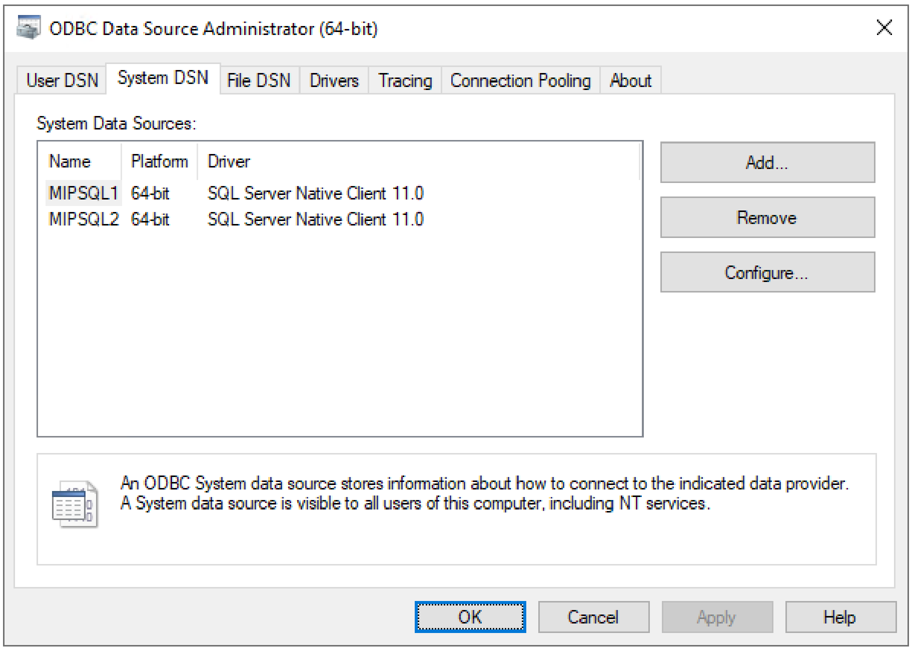


OK

Repeat these procedures for every additional database and database instance.

Use Data Source Names: **MIPSQL2, MIPSQL3, etc.**

Example for a multi system installation:



1.6.4. Remap Database User to SQL Server Login

After restoring a database “mip1” it might be necessary to remap the database user “mipadm” to the SQL Server login “mipadm” by using one of the following commands:

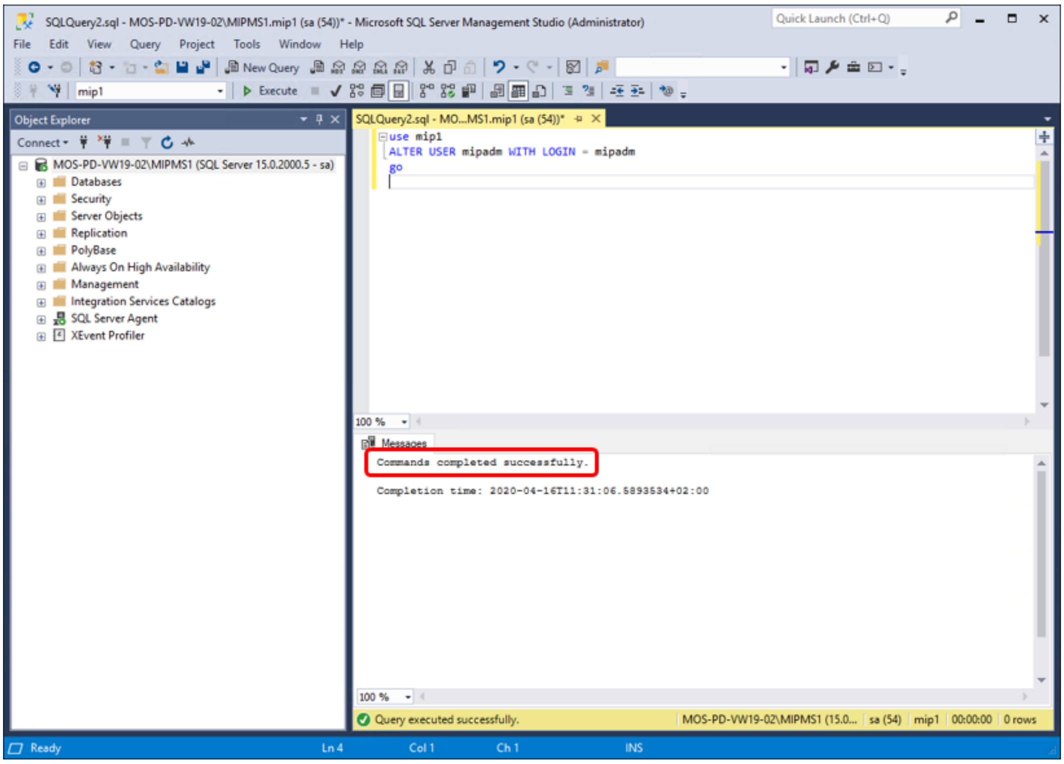
According to Microsoft the procedure “sp_change_users_login” is currently in maintenance mode and may be removed in a future version of Microsoft SQL Server.

```
sp_change_users_login 'update_one', 'username', 'loginname'
```

```
use mip1
sp_change_users_login 'update_one', 'mipadm', 'mipadm'
go
```

Microsoft recommends to use the command “ALTER USER” instead, e.g.:

```
use mip1
ALTER USER mipadm WITH LOGIN = mipadm
go
```



1.7. Separate Database and Application Server

With **separate application and database servers** for MIP based products the following software is needed on the MIP based application server and must **always** be installed on the **application server**.

The SQL Server Native Client is needed by MIP based products to connect to its database.

The following SQL Server tools are mandatory for several MIP, HYDRA or FEDRA functionality:
bcp.exe (needed by hy2exp.exe, hy2imp.exe and PDV 8.3)
sqlcmd.exe (needed by hy_diff_mw4.scr)

Please make always sure that the SQL Server tools **bcp.exe** and **sqlcmd.exe** are available on the **application server** of your MIP based product!

Those tools are part of the "SQL Server Management Studio" installation package as long as the version of your SSMS is not higher than version 17.

If using a SSMS version >17 you need to install the 64-Bit version of the "**Microsoft Command Line Utilities 14.0 for SQL Server**", e.g. to be download from:
https://www.microsoft.com/en-us/download/details.aspx?id=53591

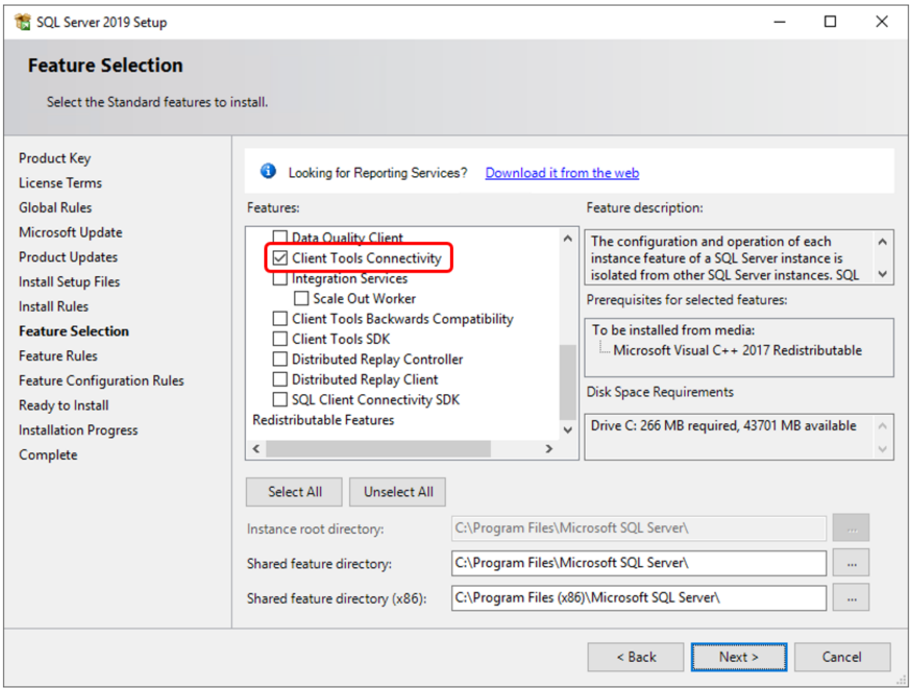
The Microsoft Command Line Utilities requires an already installed 64-Bit version of the "**Microsoft ODBC Driver 13.1 for SQL Server**", e.g. to be download from:
https://www.microsoft.com/en-us/download/details.aspx?id=53339

Proceed with the installation as described below.

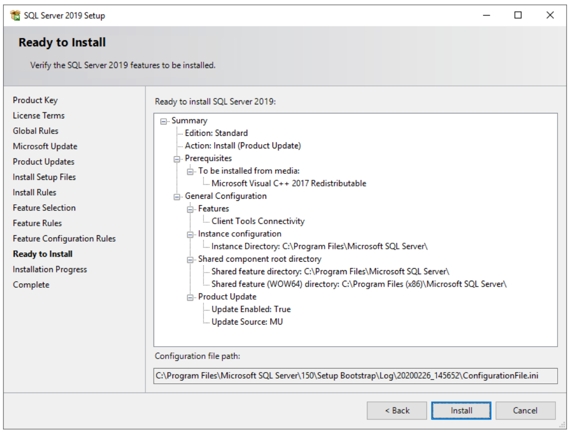
1.7.1. Installing SQL Server Client

On the **application server** of your MIP based product start the installation as described in chapter "1.2. Installing SQL Server 2019".

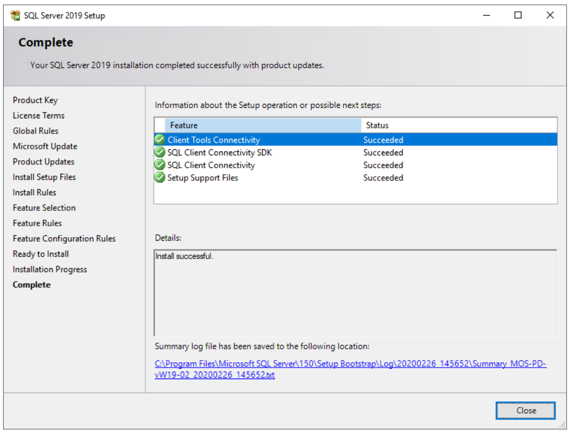
Choose the feature "**Client Tools Connectivity**" to install:



Next



Install



Close

1.7.2. ODBC Configuration

Always perform the ODBC configuration on the **application server** of your MIP based product as described in chapter "1.6.3. ODBC Configuration".

1.7.3. Installing SQL Server Management Studio

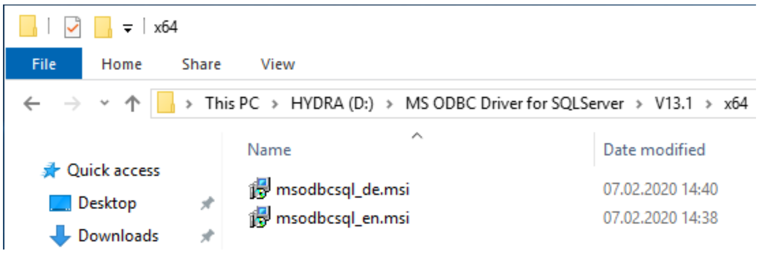
Always install the "SQL Server Management Studio" on the **application server** of your MIP based product as described in chapter "1.3. Installing SQL Server Management Studio".

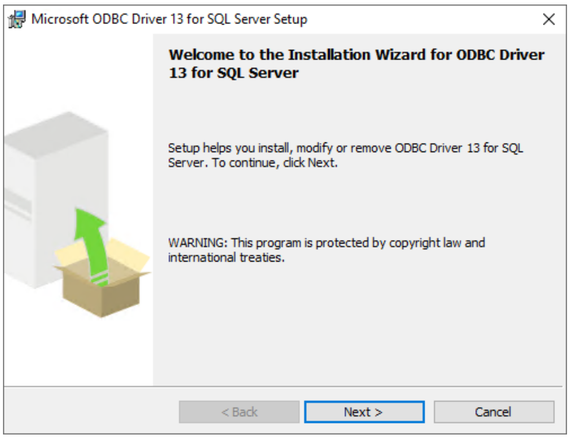
If the version of your SQL Server Management Studio is **higher than version 17** you need to proceed with the following chapters.

1.7.4. Installing Microsoft ODBC Driver 13.1

The Microsoft Command Line Utilities require an already installed 64-Bit version of the "**Microsoft ODBC Driver 13.1 for SQL Server**", e.g. to be download from:
https://www.microsoft.com/en-us/download/details.aspx?id=53339

Depending on the language setting of your application server execute one of the following files: **msodbcsql_de.msi** or **msodbcsql_en.msi**.





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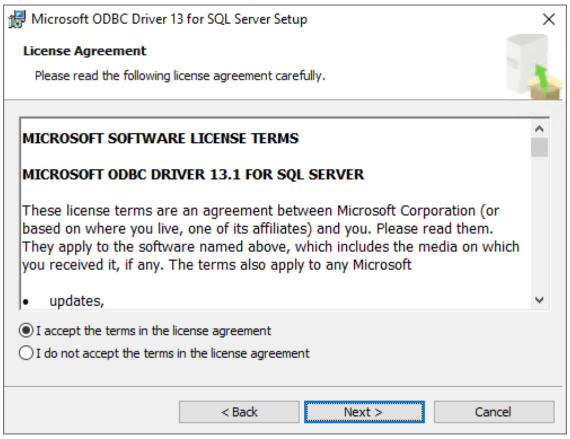
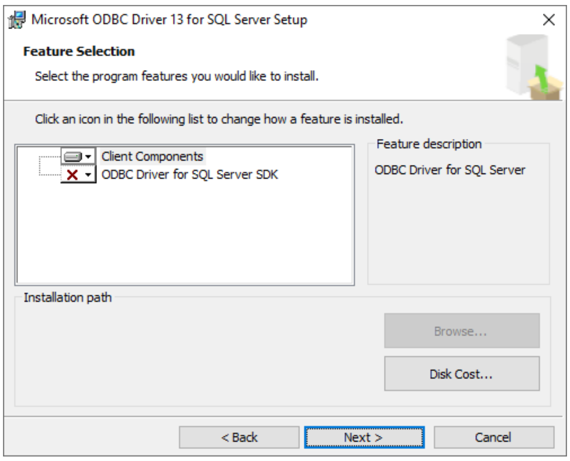
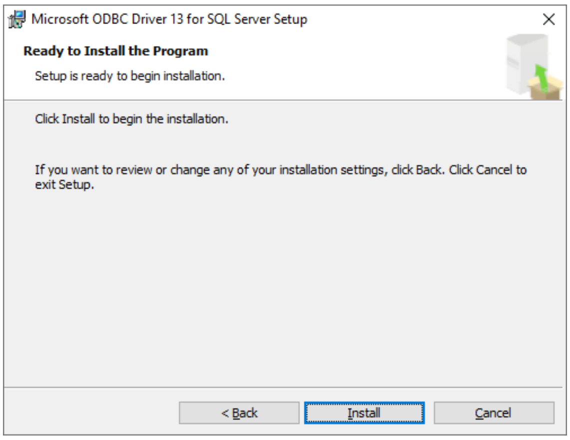


I accept the terms in the license agreement

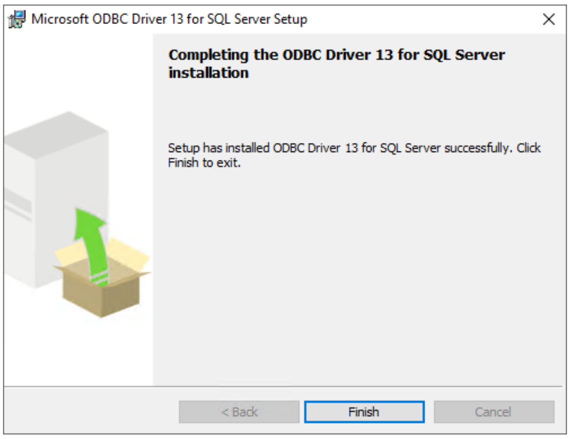
Next



Next



Install

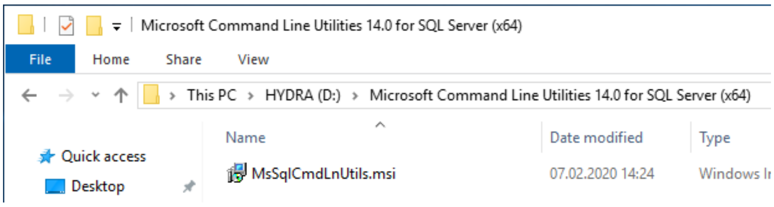

Microsoft ODBC Driver 13 for SQL Server Setup
Completing the ODBC Driver 13 for SQL Server installation
Setup has installed ODBC Driver 13 for SQL Server successfully. Click Finish to exit.
< Back
Finish
Cancel

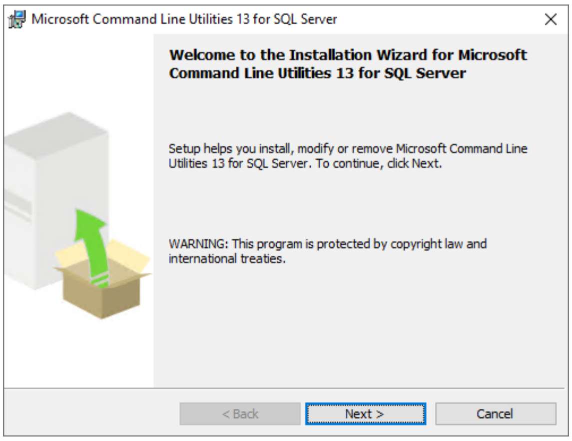
Finish

1.7.5. Installing Microsoft Command Line Utilities

If using a SSMS version >17 you need to install the 64-Bit version of the "**Microsoft Command Line Utilities 14.0 for SQL Server**", e.g. to be download from:
https://www.microsoft.com/en-us/download/details.aspx?id=53591

Execute the file **MsSqlCmdLnUtils.msi**.





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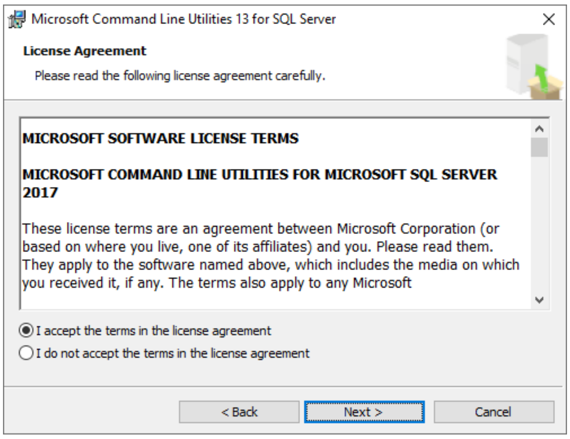
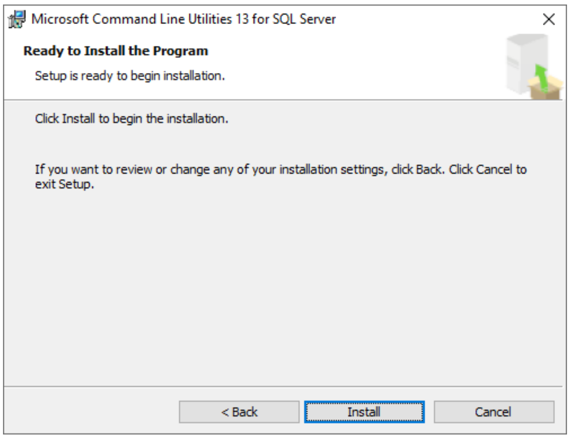
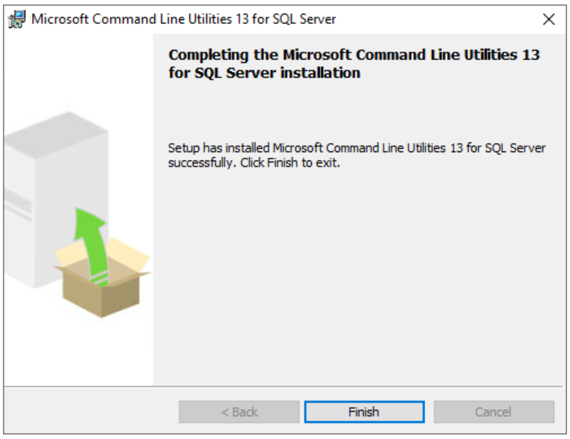


I accept the terms in the license agreement

Next



Install



Finish

Please make always sure that the SQL Server tools **bcp.exe** and **sqlcmd.exe** are available on the **application server** of your MIP based product!

bcp.exe (needed by hy2exp.exe, hy2imp.exe and PDV 8.3) should now be available, e.g. in:

`c:\Program Files\Microsoft SQL Server\Client SDK\ODBC\130\Tools\Binn\bcp.exe`

sqlcmd.exe (needed by hy_diff_mw4.scr) should now be available, e.g. in:

`c:\Program Files\Microsoft SQL Server\Client SDK\ODBC\130\Tools\Binn\SQLCMD.EXE`